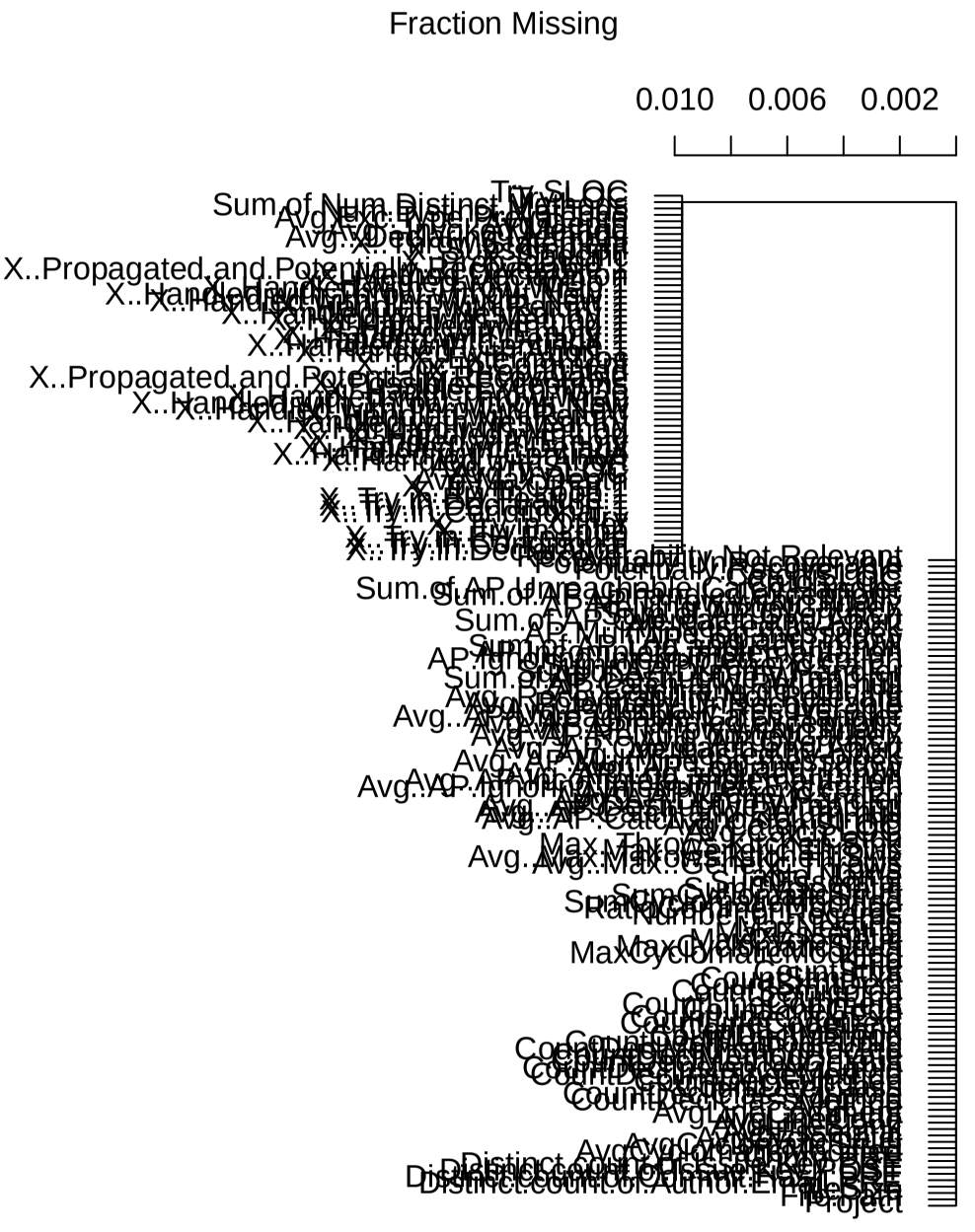
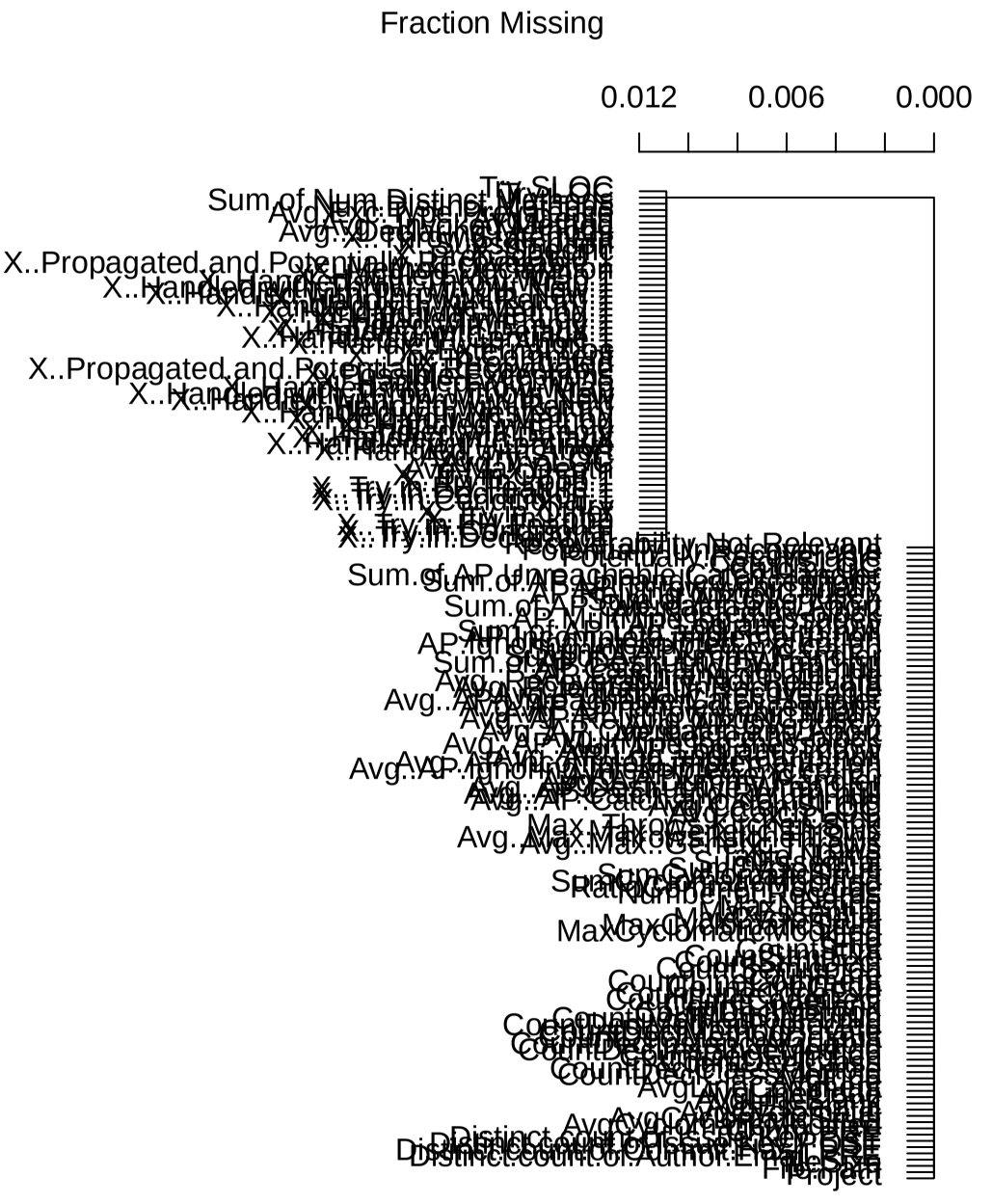
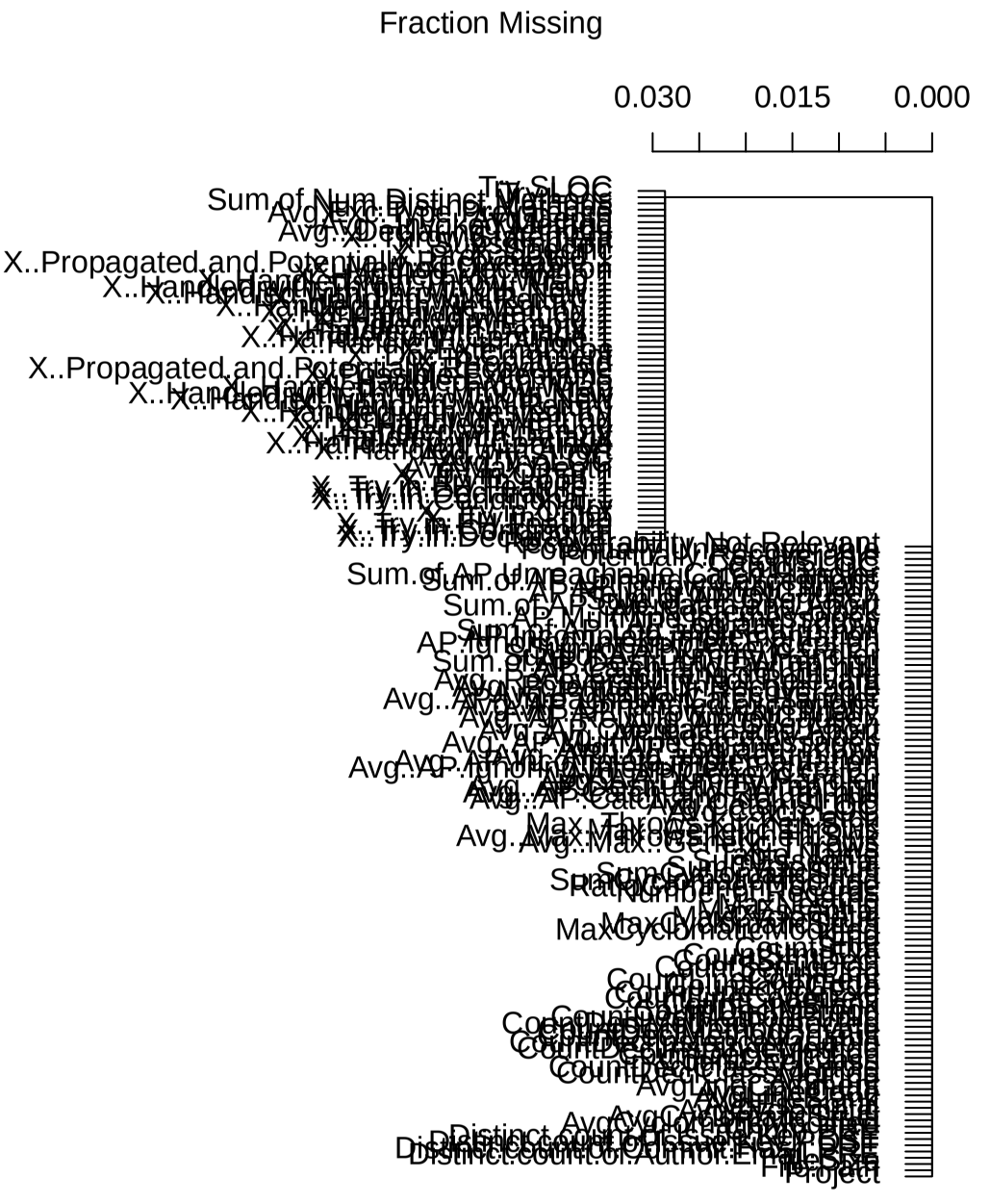
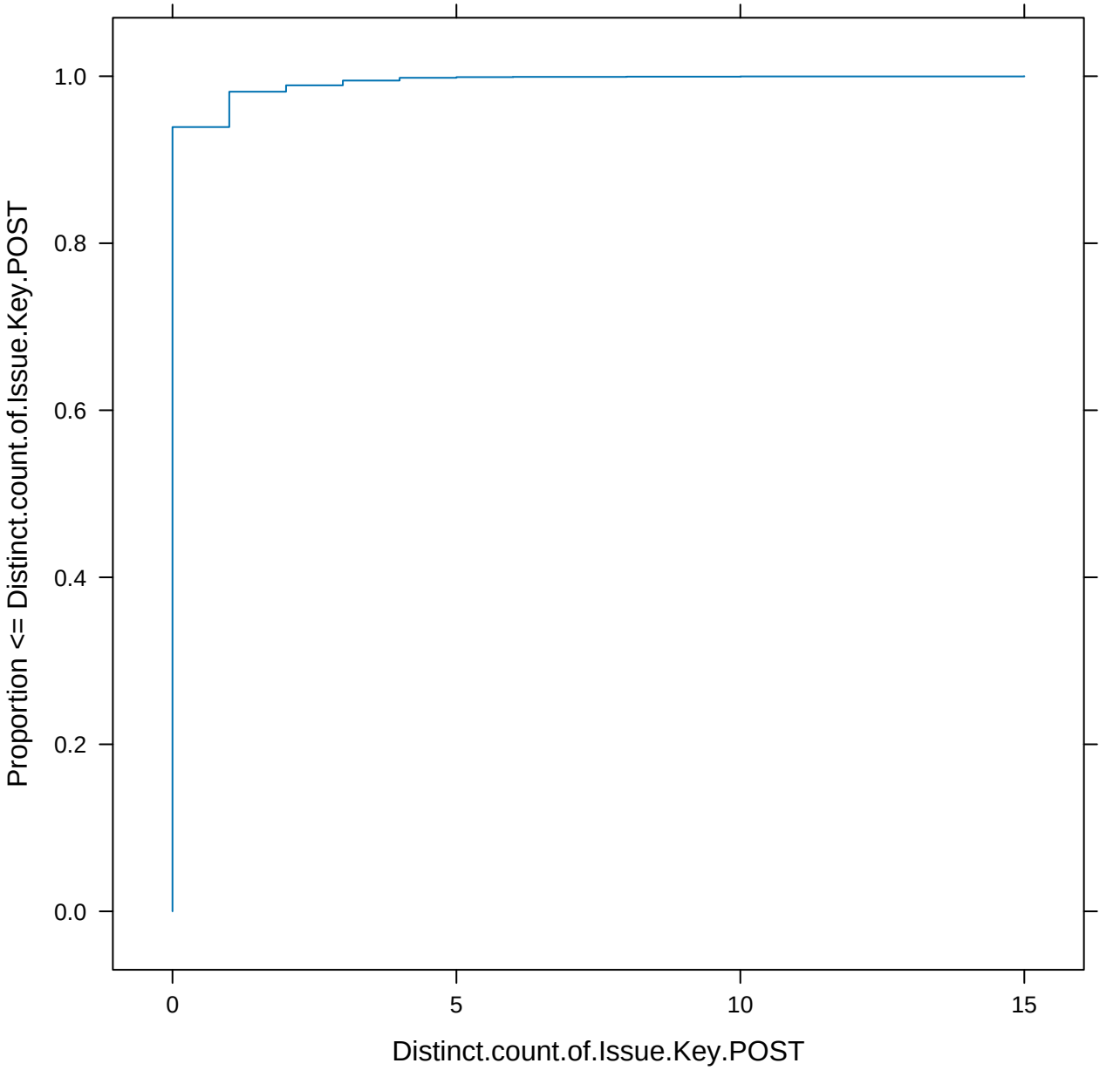
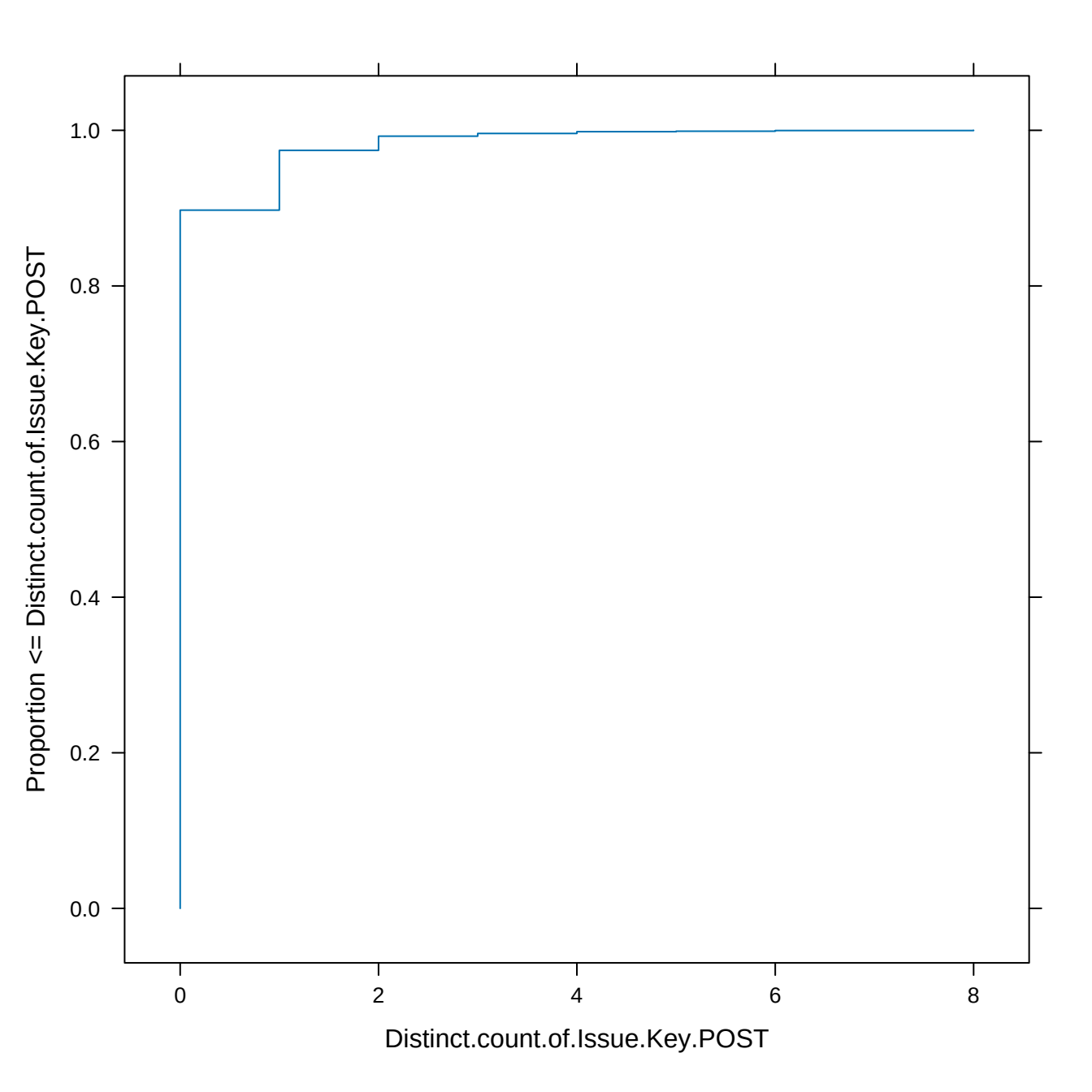
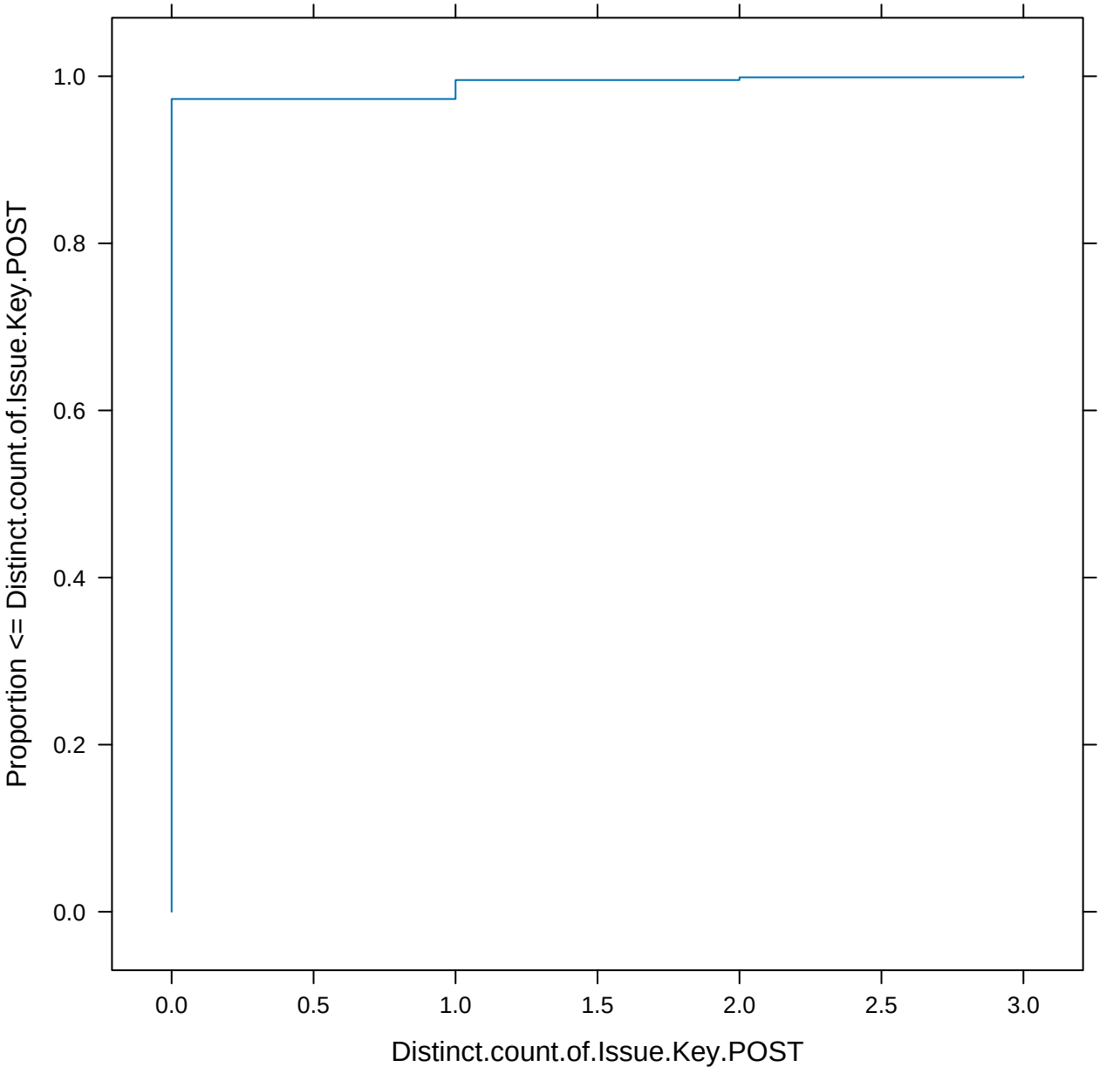

Fraction Missing
0.010
0.006
0.002
Try.SLOC
Sum.of.Num.Distinct.Methods
Sum.of.AP.Unreachable.Calls
Project

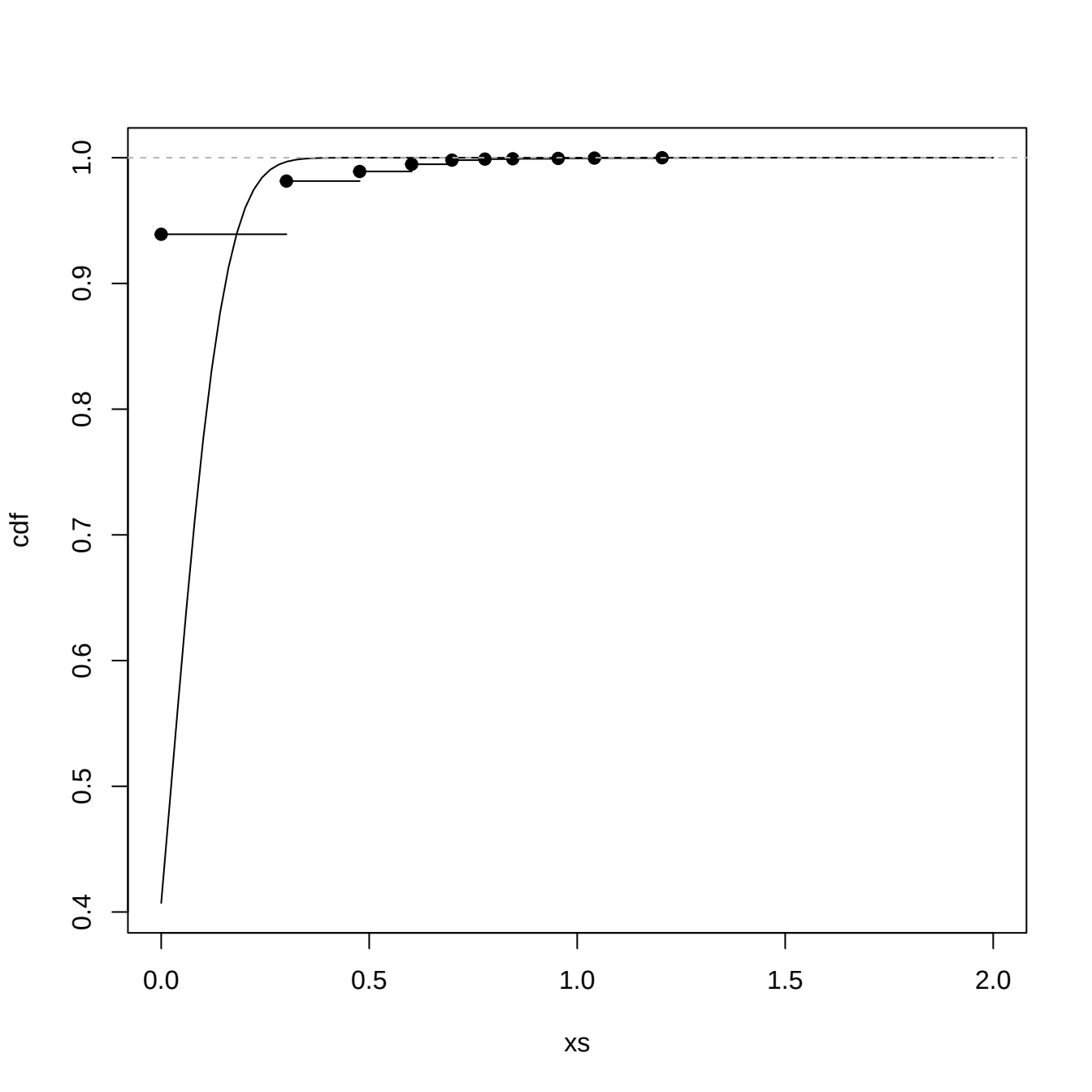

Fraction Missing
0.012
0.006
0.000
Try.SLOC
Project

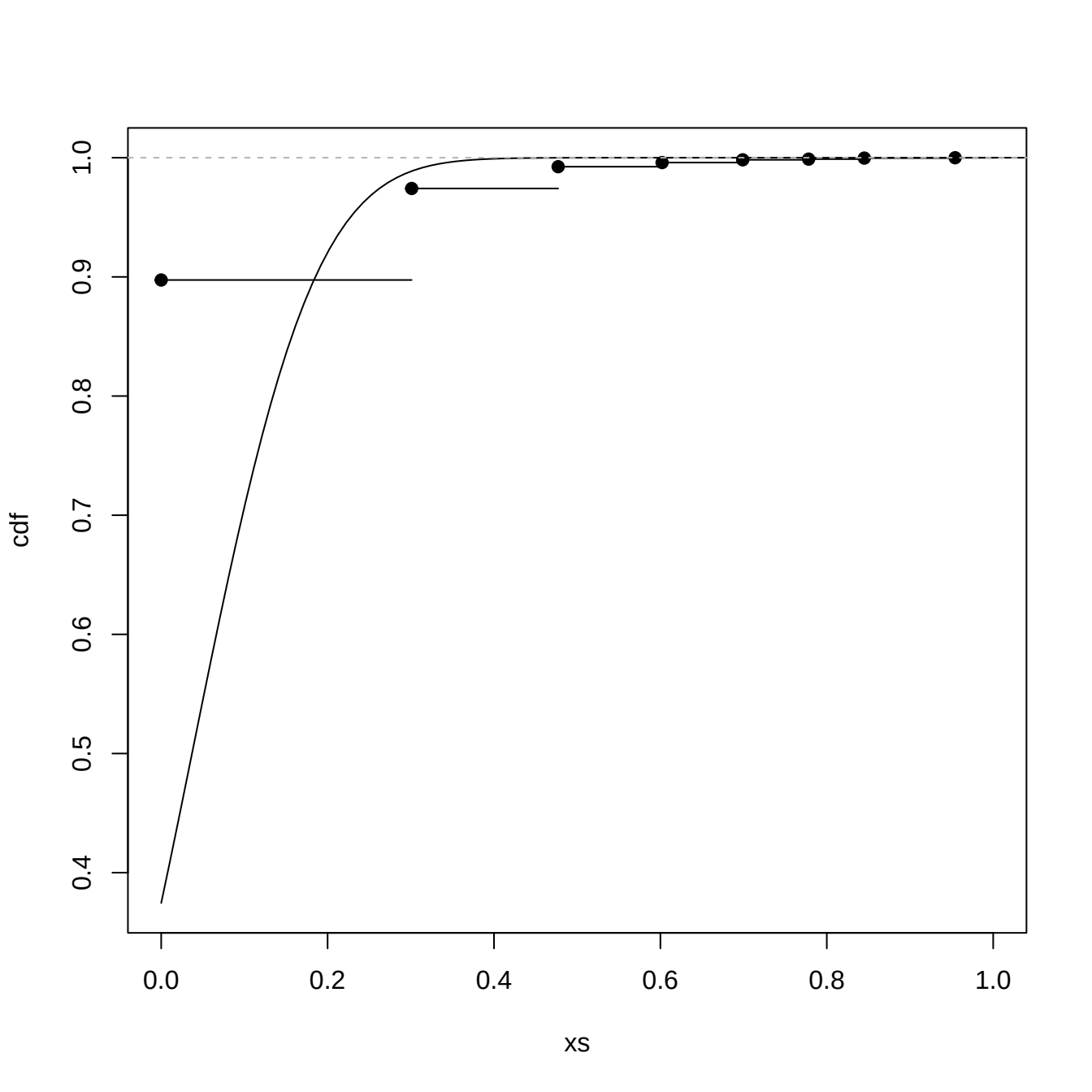

Fraction Missing
0.030
0.015
0.000
Try.SLOC
Sum.of.Num.Distinct.Methods
Sum.of.AP.Unreachable.Calls
Project

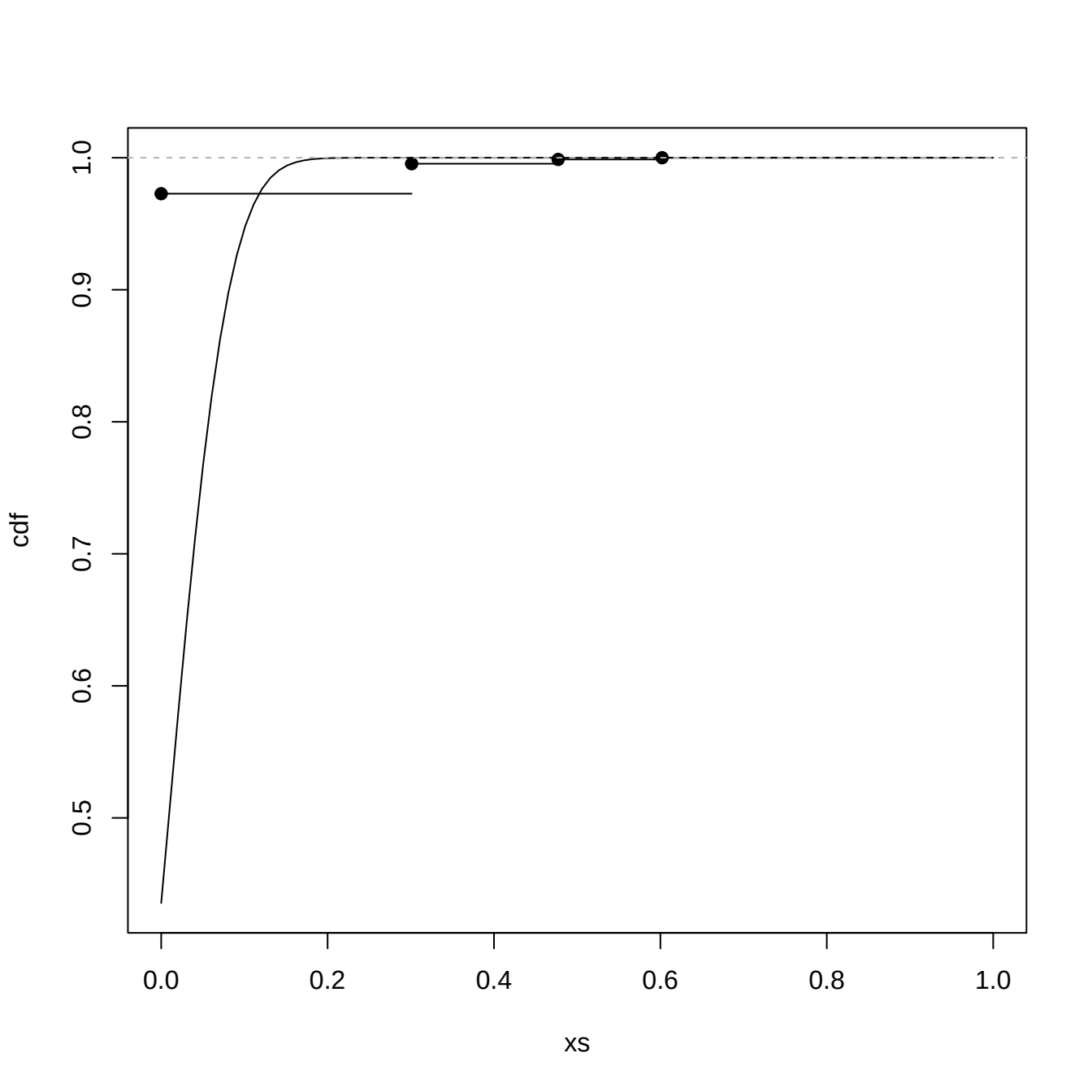

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
Distinct.count.of.Issue.Key.POST

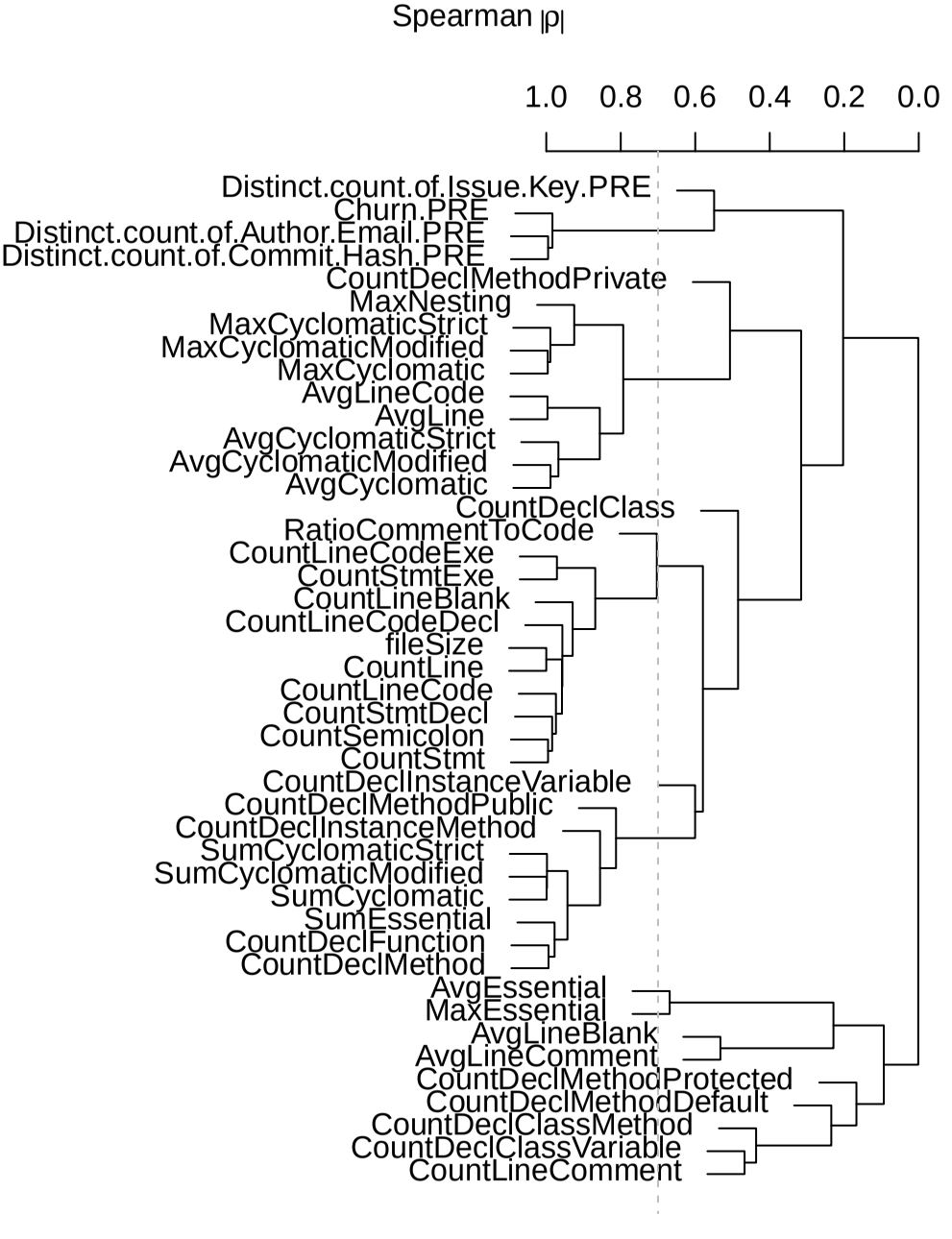

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Distinct.count.of.Issue.Key.POST

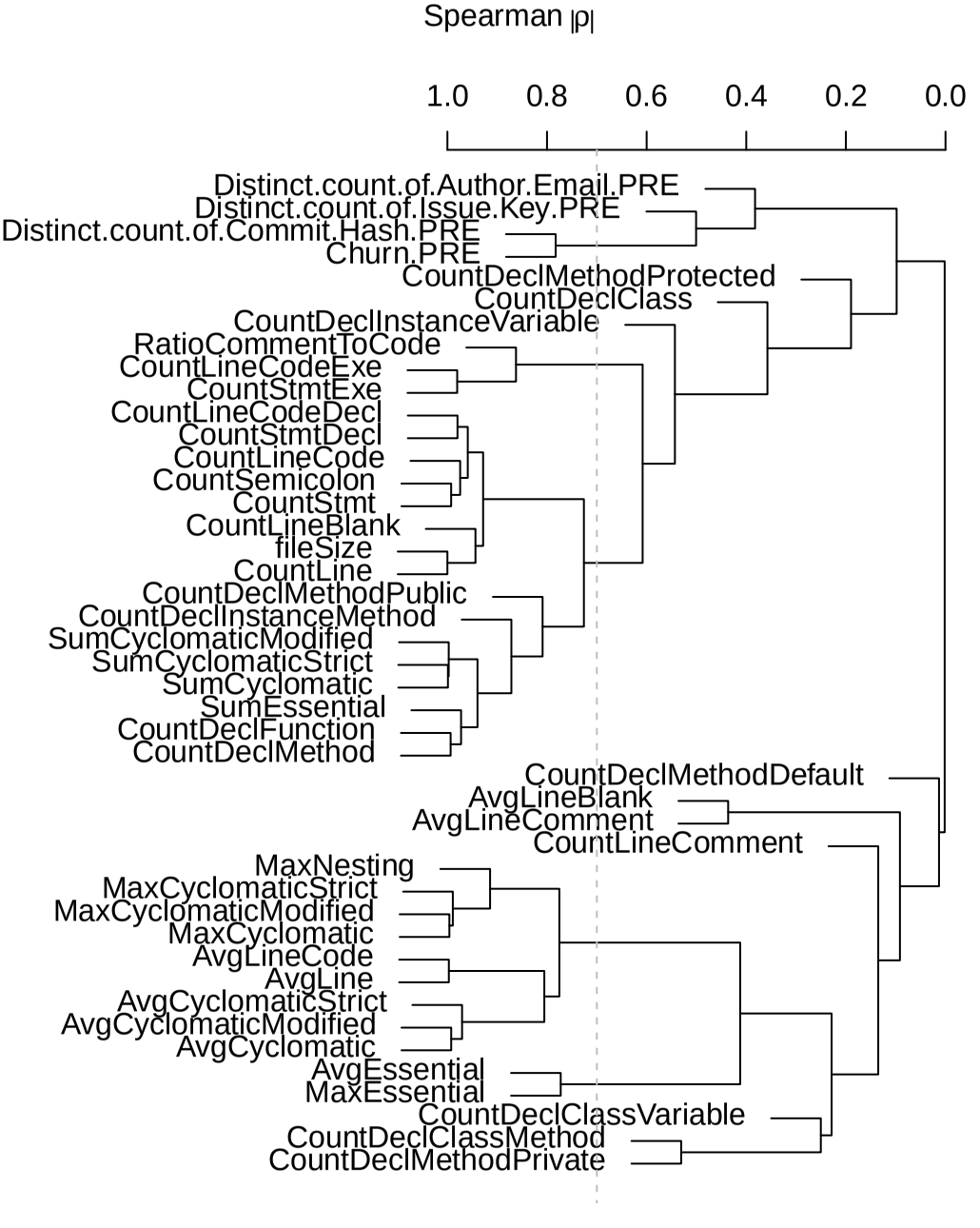

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Distinct.count.of.Issue.Key.POST

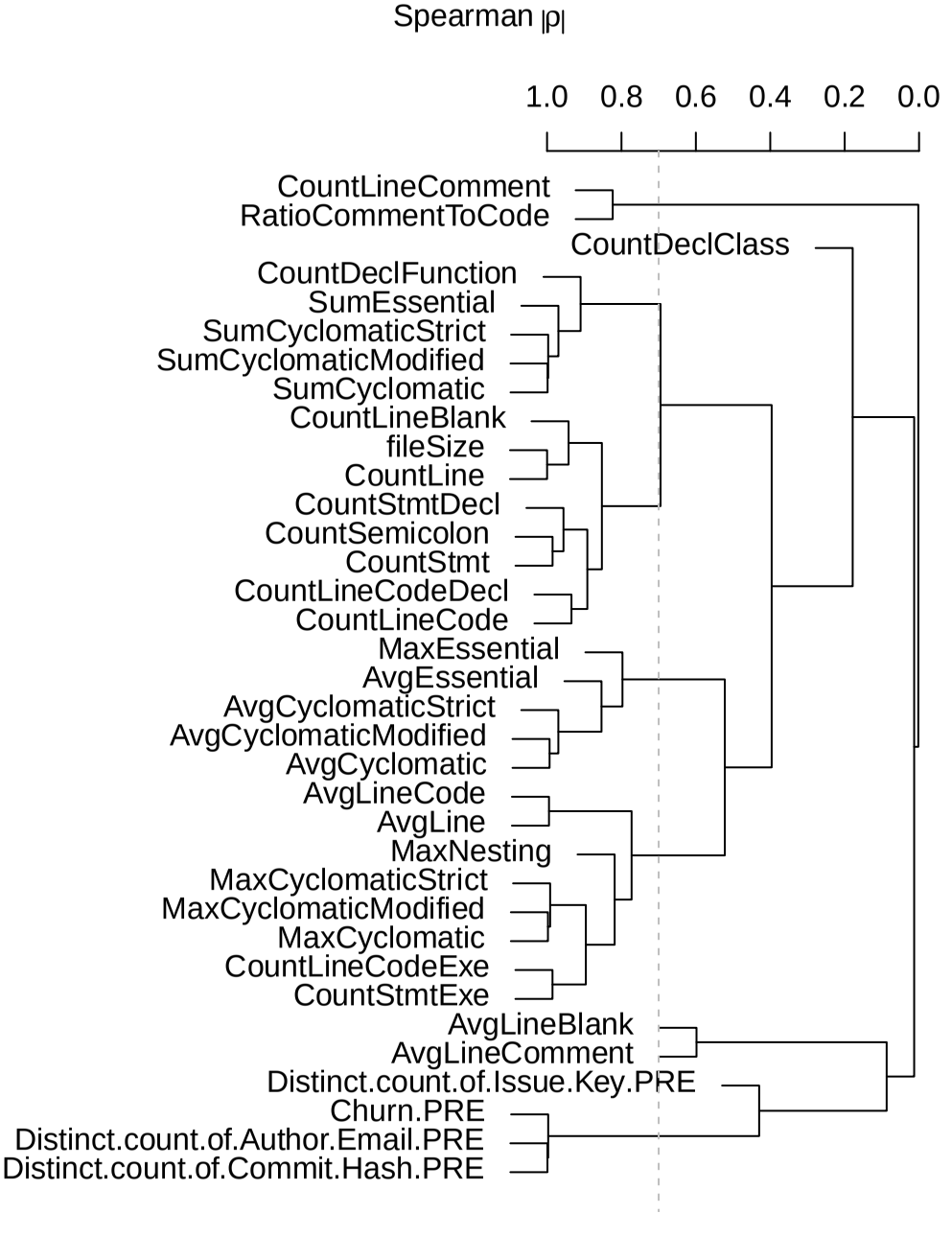

cdf
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.0
0.5
1.0
1.5
2.0
xs

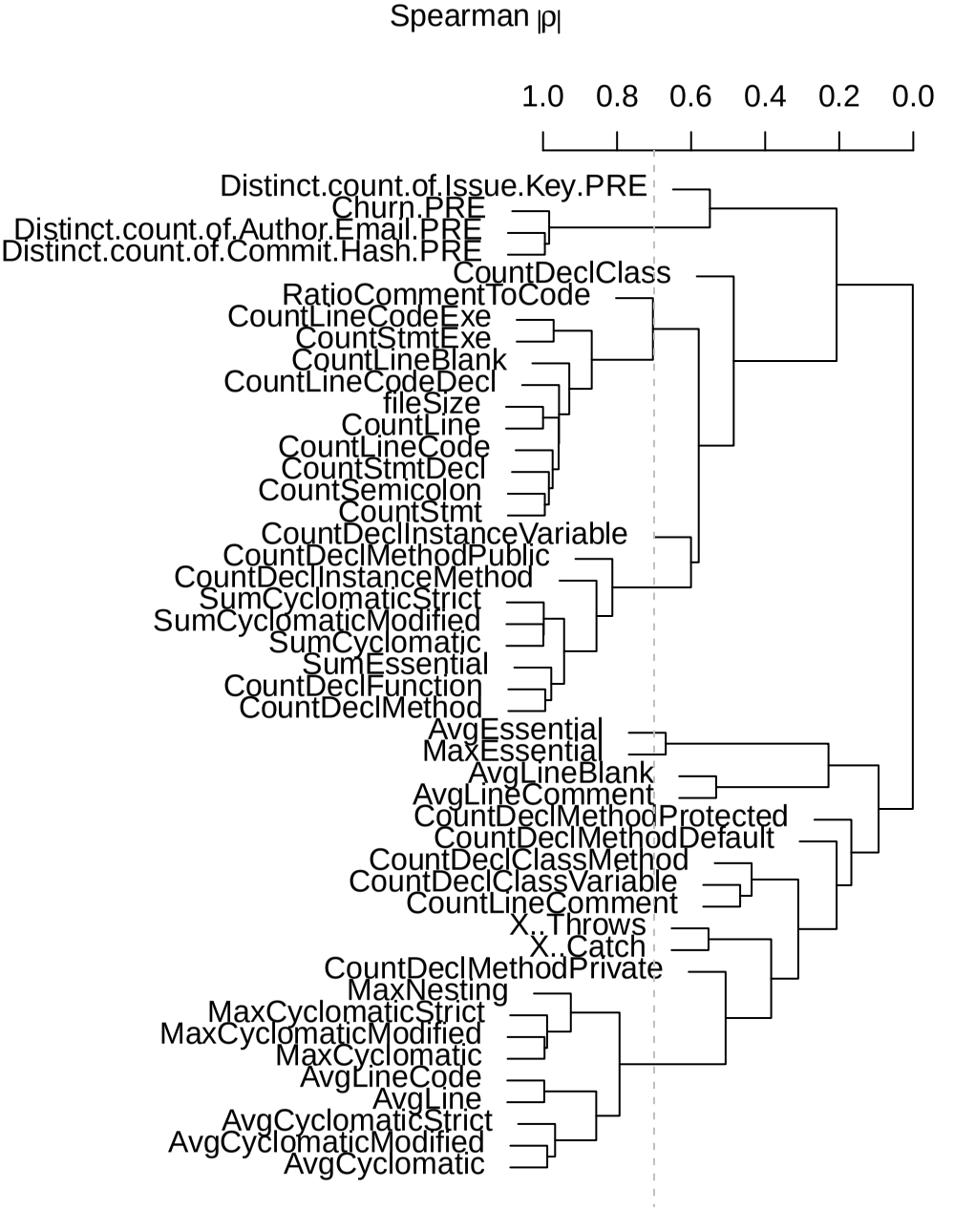

cdf
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.0
0.2
0.4
0.6
0.8
1.0
xs

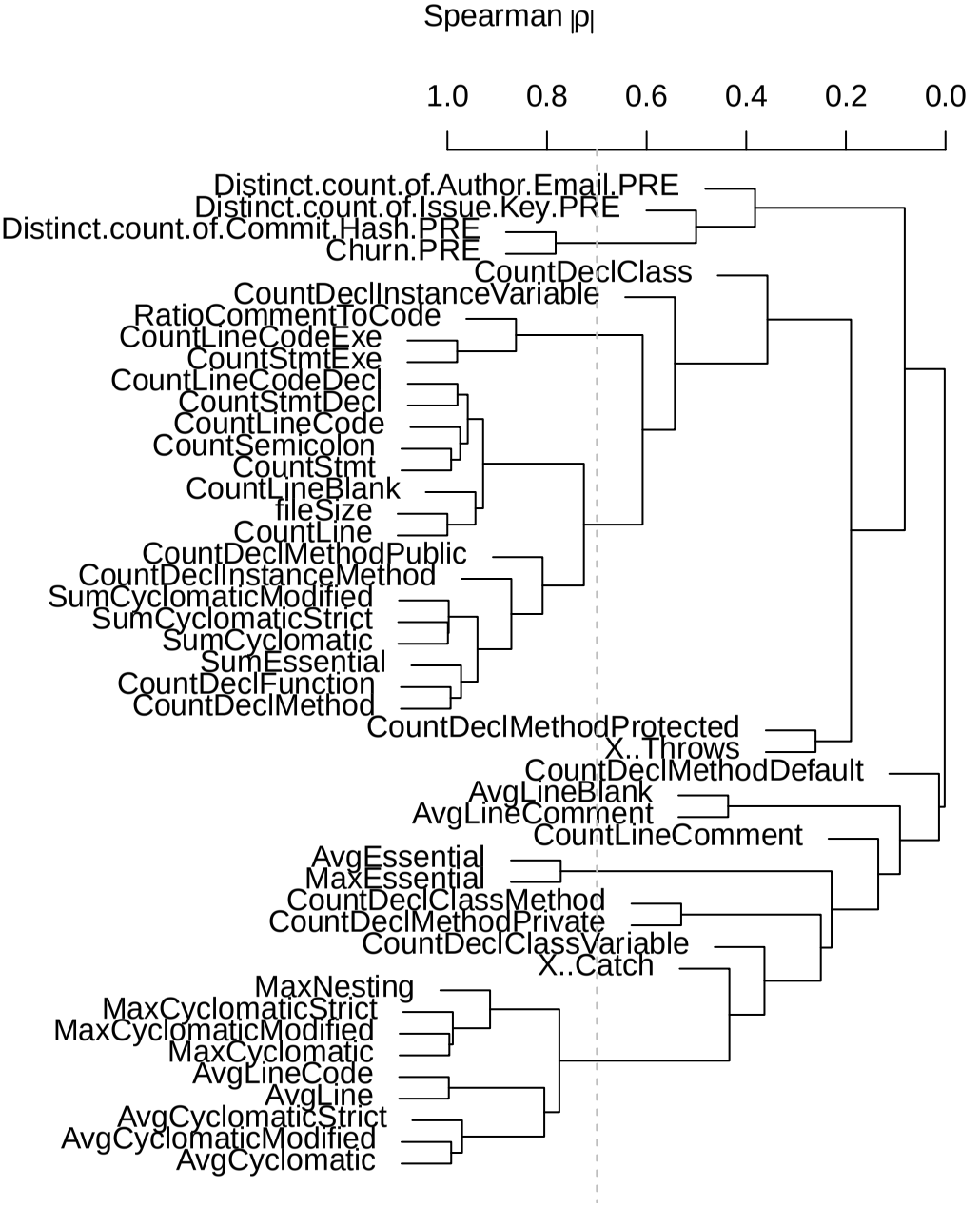

cdf
1.0
0.9
0.8
0.7
0.6
0.5
0.0
0.2
0.4
0.6
0.8
1.0
xs

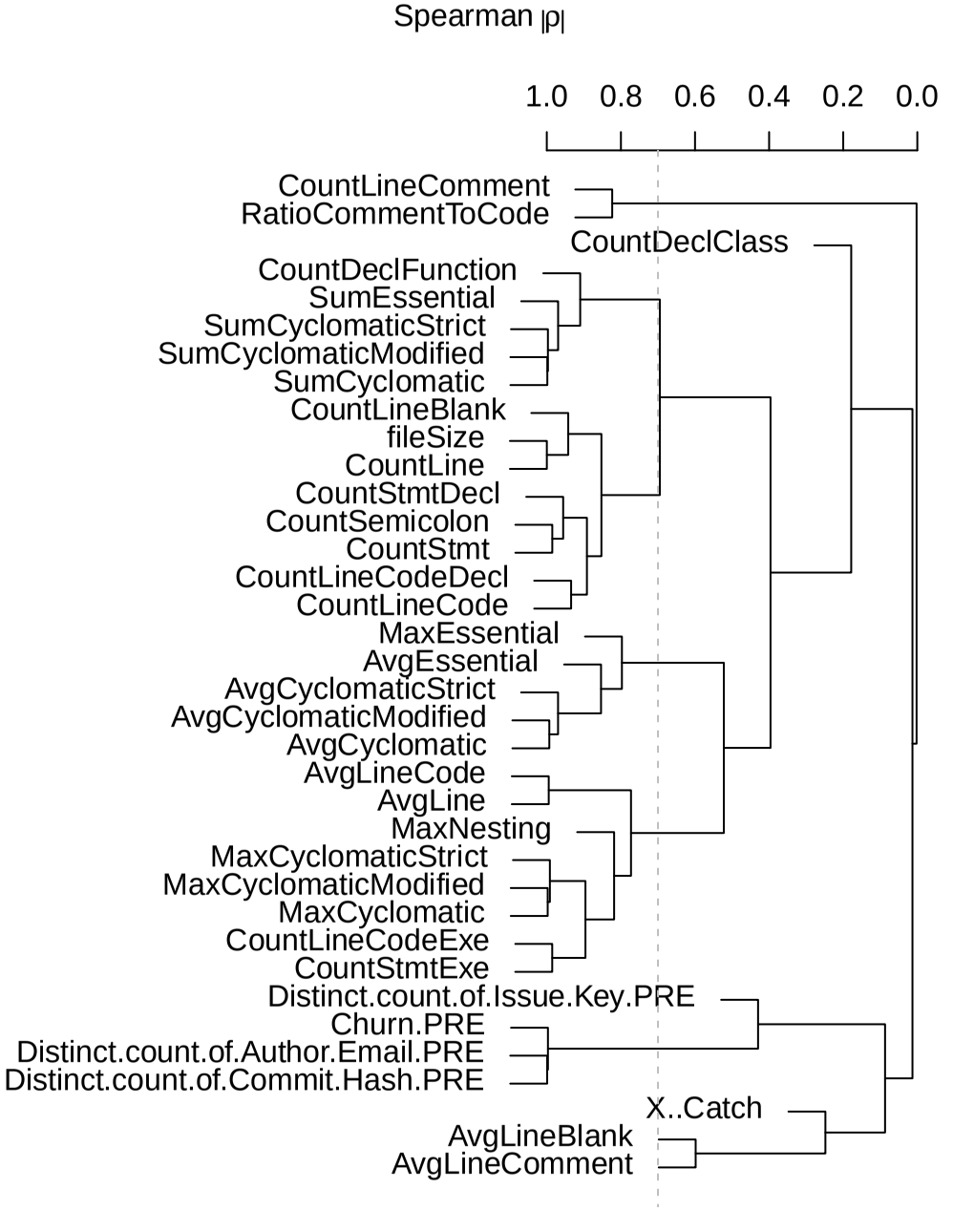

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
CountDeclMethodPrivate
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
CountDeclClass
RatioCommentToCode
CountLineCodeExe
CountStmtExe
CountLineBlank
CountLineCodeDecl
fileSize
CountLine
CountLineCode
CountStmtDecl
CountSemicolon
CountStmt
CountDeclInstanceVariable
CountDeclMethodPublic
CountDeclInstanceMethod
SumCyclomaticStrict
SumCyclomaticModified
SumCyclomatic
SumEssential
CountDeclFunction
CountDeclMethod
AvgEssential
MaxEssential
AvgLineBlank
AvgLineComment
CountDeclMethodProtected
CountDeclMethodDefault
CountDeclClassMethod
CountDeclClassVariable
CountLineComment

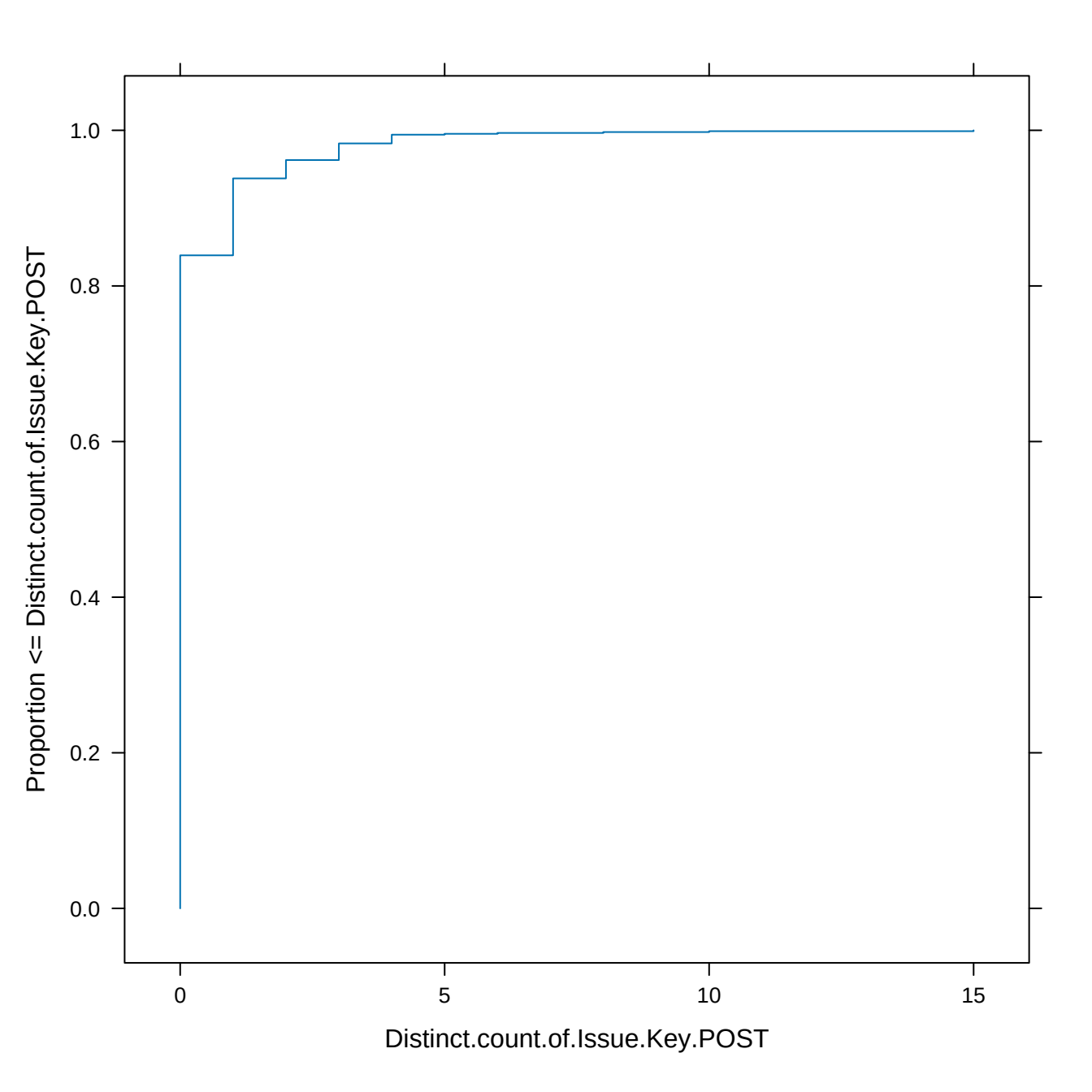

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Issue.Key.PRE
Distinct.count.of.Commit.Hash.PRE
Churn.PRE
CountDeclMethodProtected
CountDeclClass
CountDeclInstanceVariable
RatioCommentToCode
CountLineCodeExe
CountStmtExe
CountLineCodeDecl
CountStmtDecl
CountLineCode
CountSemicolon
CountStmt
CountLineBlank
fileSize
CountLine
CountDeclMethodPublic
CountDeclInstanceMethod
SumCyclomaticModified
SumCyclomaticStrict
SumCyclomatic
SumEssential
CountDeclFunction
CountDeclMethod
CountDeclMethodDefault
AvgLineBlank
AvgLineComment
CountLineComment
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
AvgEssential
MaxEssential
CountDeclClassVariable
CountDeclClassMethod
CountDeclMethodPrivate

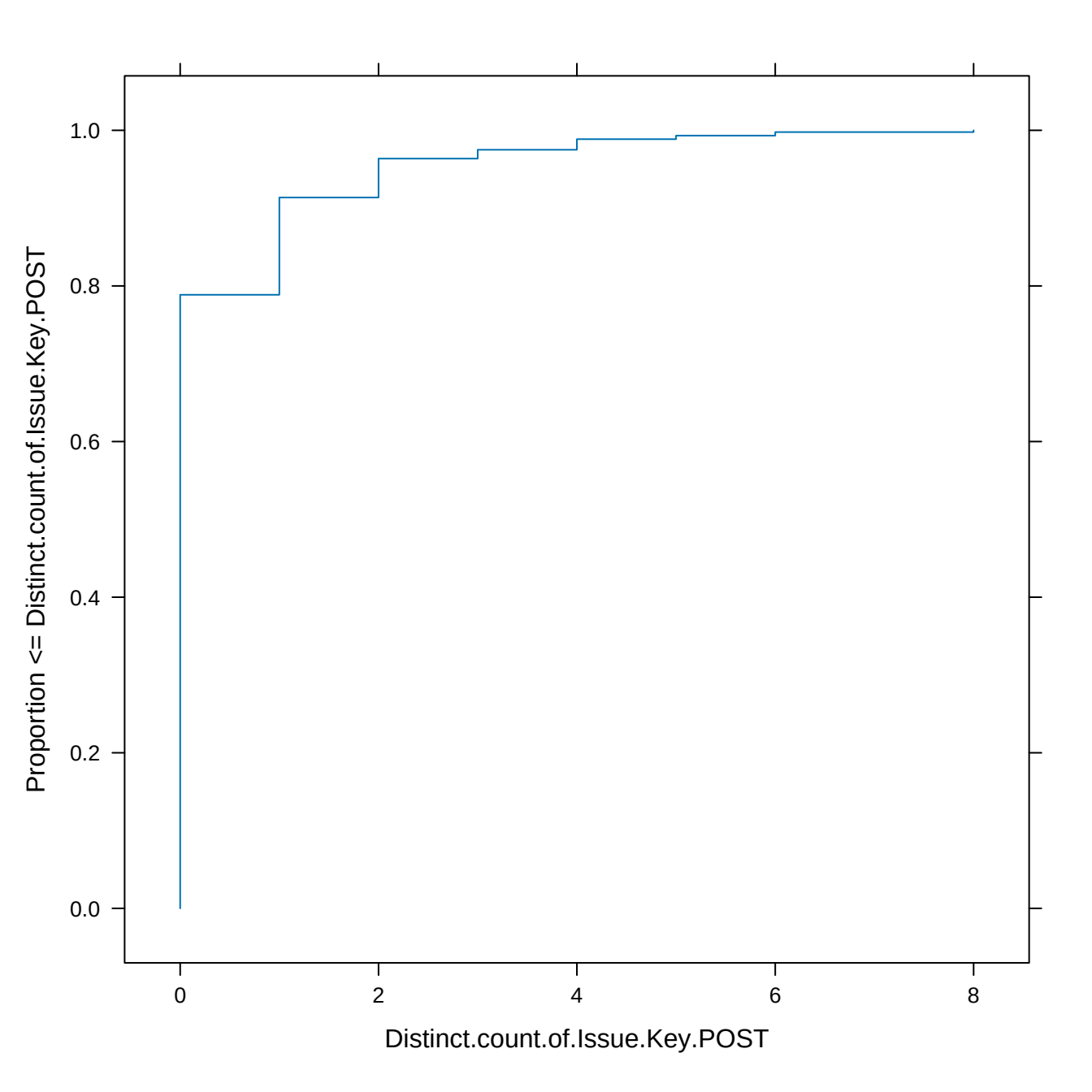

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
CountLineComment
RatioCommentToCode
CountDeclClass
CountDeclFunction
SumEssential
SumCyclomaticStrict
SumCyclomaticModified
SumCyclomatic
CountLineBlank
fileSize
CountLine
CountStmtDecl
CountSemicolon
CountStmt
CountLineCodeDecl
CountLineCode
MaxEssential
AvgEssential
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
AvgLineCode
AvgLine
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
CountLineCodeExe
CountStmtExe
AvgLineBlank
AvgLineComment
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE

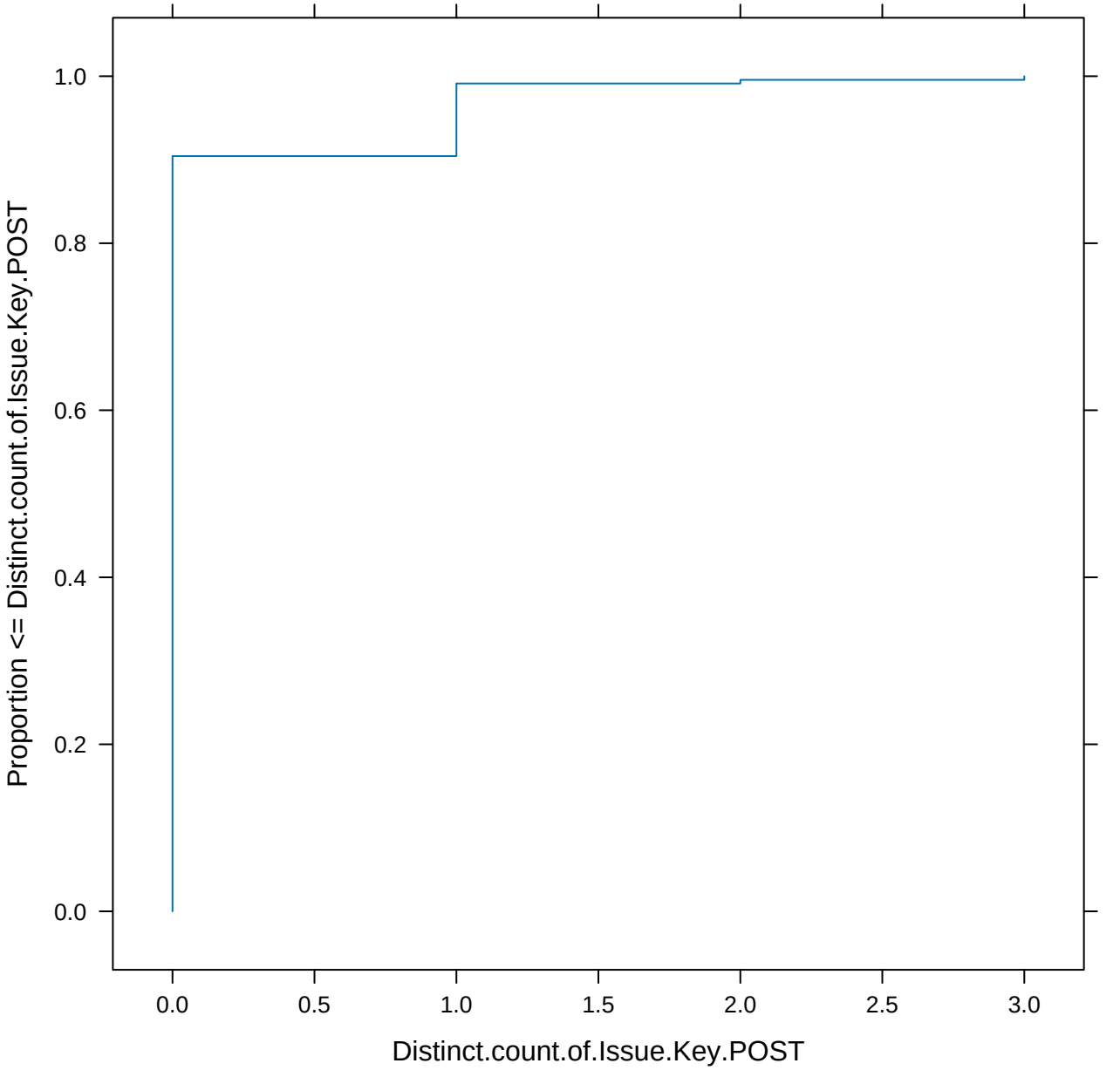

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
CountDeclClass
RatioCommentToCode
CountLineCodeExe
CountStmtExe
CountLineBlank
CountLineCodeDecl
fileSize
CountLine
CountLineCode
CountStmtDecl
CountSemicolon
CountStmt
CountDeclInstanceVariable
CountDeclMethodPublic
CountDeclInstanceMethod
SumCyclomaticStrict
SumCyclomaticModified
SumCyclomatic
SumEssential
CountDeclFunction
CountDeclMethod
AvgEssential
MaxEssential
AvgLineBlank
AvgLineComment
CountDeclMethodProtected
CountDeclMethodDefault
CountDeclClassMethod
CountDeclClassVariable
CountLineComment
X..Throws
X..Catch
CountDeclMethodPrivate
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic

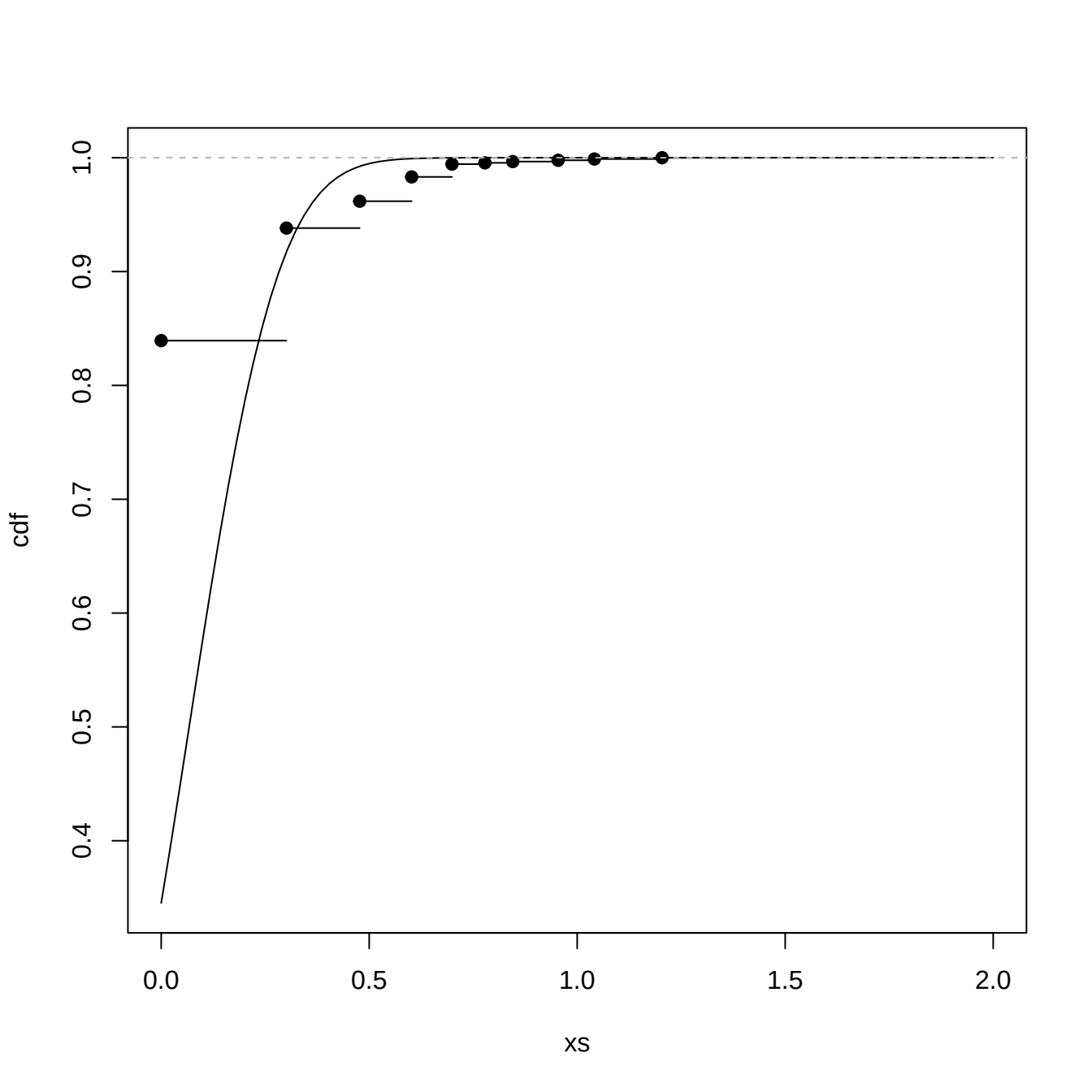

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Issue.Key.PRE
Distinct.count.of.Commit.Hash.PRE
Churn.PRE
CountDeclClass
CountDeclInstanceVariable
RatioCommentToCode
CountLineCodeExe
CountStmtExe
CountLineCodeDecl
CountStmtDecl
CountLineCode
CountSemicolon
CountStmt
CountLineBlank
fileSize
CountLine
CountDeclMethodPublic
CountDeclInstanceMethod
SumCyclomaticModified
SumCyclomaticStrict
SumCyclomatic
SumEssential
CountDeclFunction
CountDeclMethod
CountDeclMethodProtected
X..Throws
CountDeclMethodDefault
AvgLineBlank
AvgLineComment
CountLineComment
AvgEssential
MaxEssential
CountDeclClassMethod
CountDeclMethodPrivate
CountDeclClassVariable
X..Catch
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic

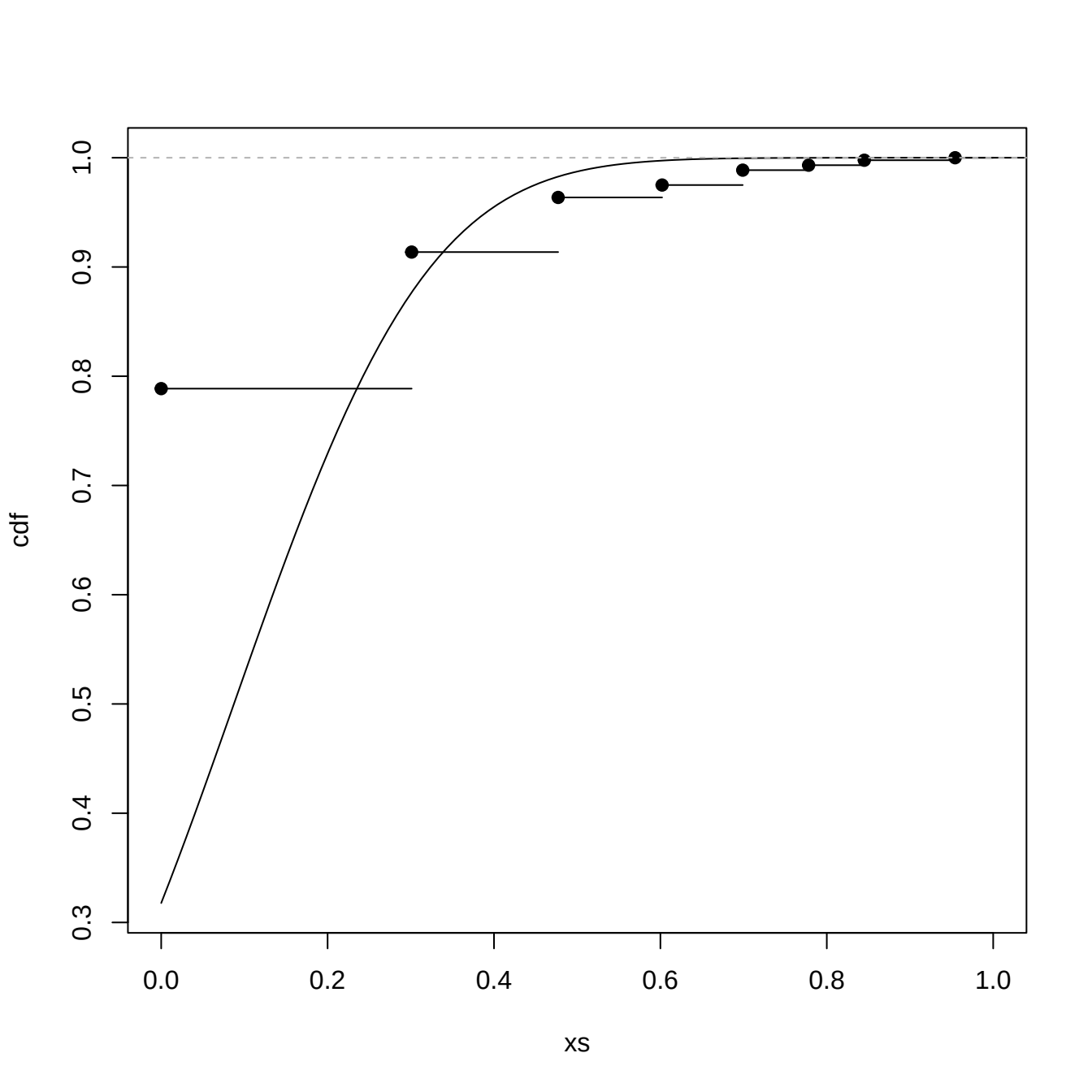

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
CountLineComment
RatioCommentToCode
CountDeclClass
CountDeclFunction
SumEssential
SumCyclomaticStrict
SumCyclomaticModified
SumCyclomatic
CountLineBlank
fileSize
CountLine
CountStmtDecl
CountSemicolon
CountStmt
CountLineCodeDecl
CountLineCode
MaxEssential
AvgEssential
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
AvgLineCode
AvgLine
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
CountLineCodeExe
CountStmtExe
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
X..Catch
AvgLineBlank
AvgLineComment

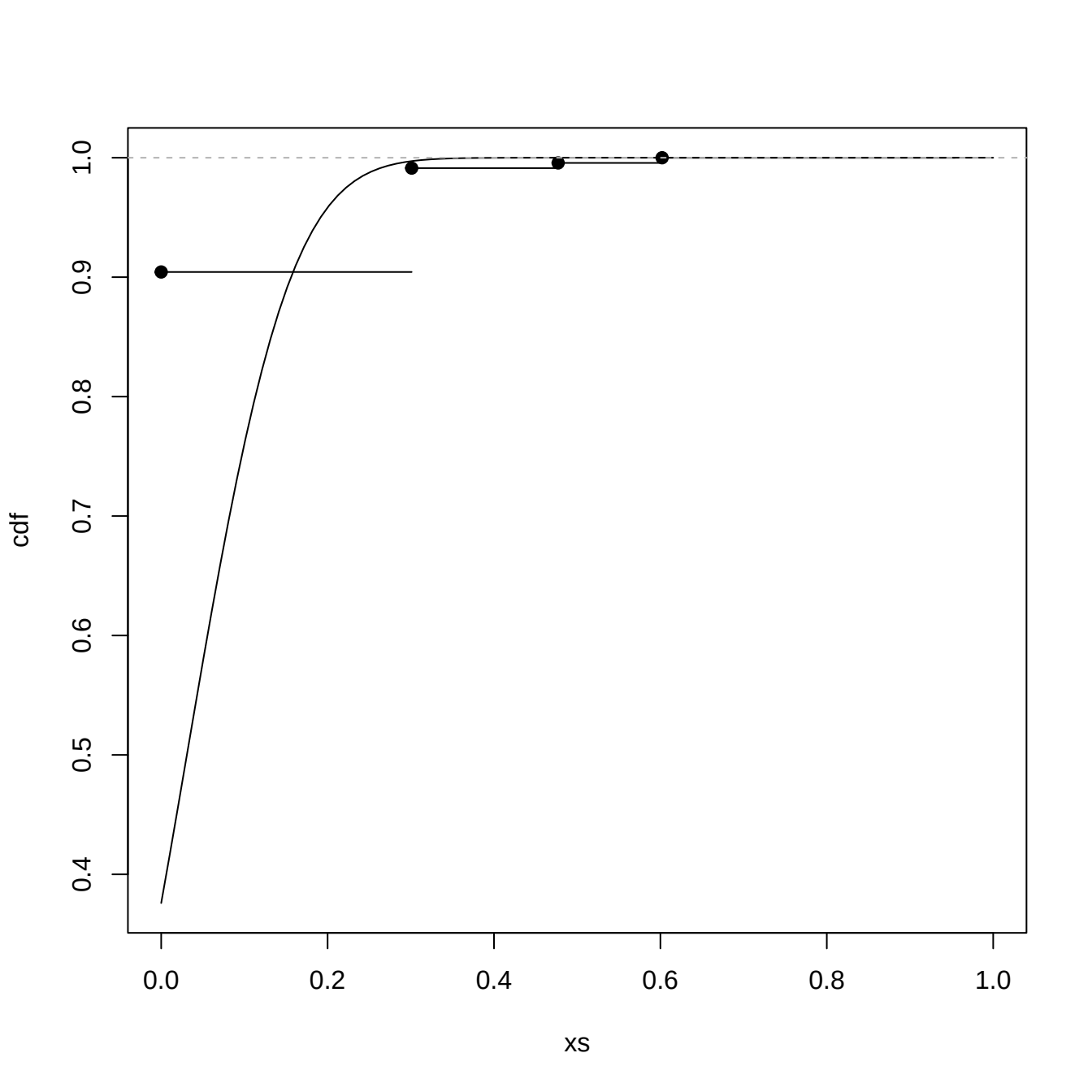

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
Distinct.count.of.Issue.Key.POST

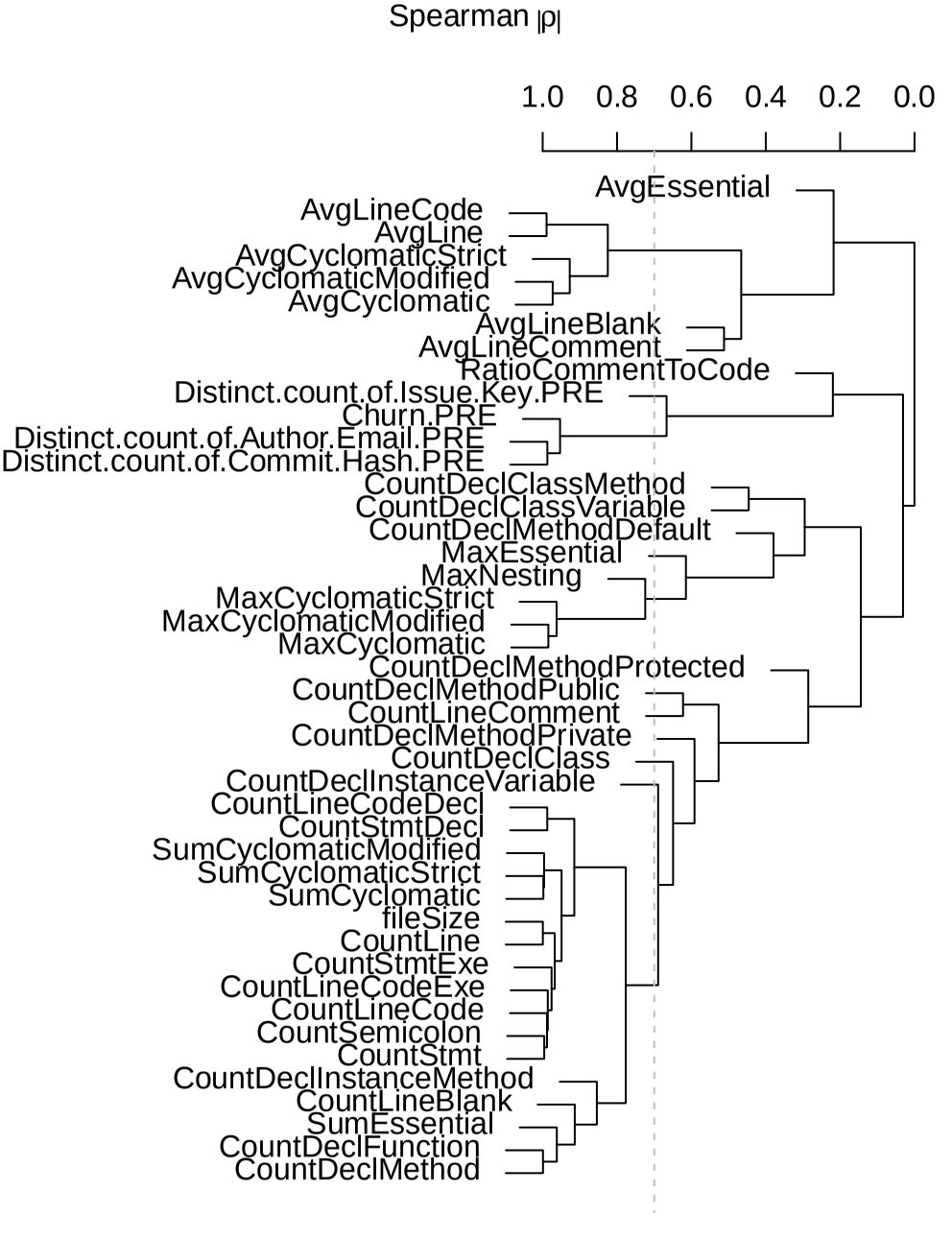

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Distinct.count.of.Issue.Key.POST

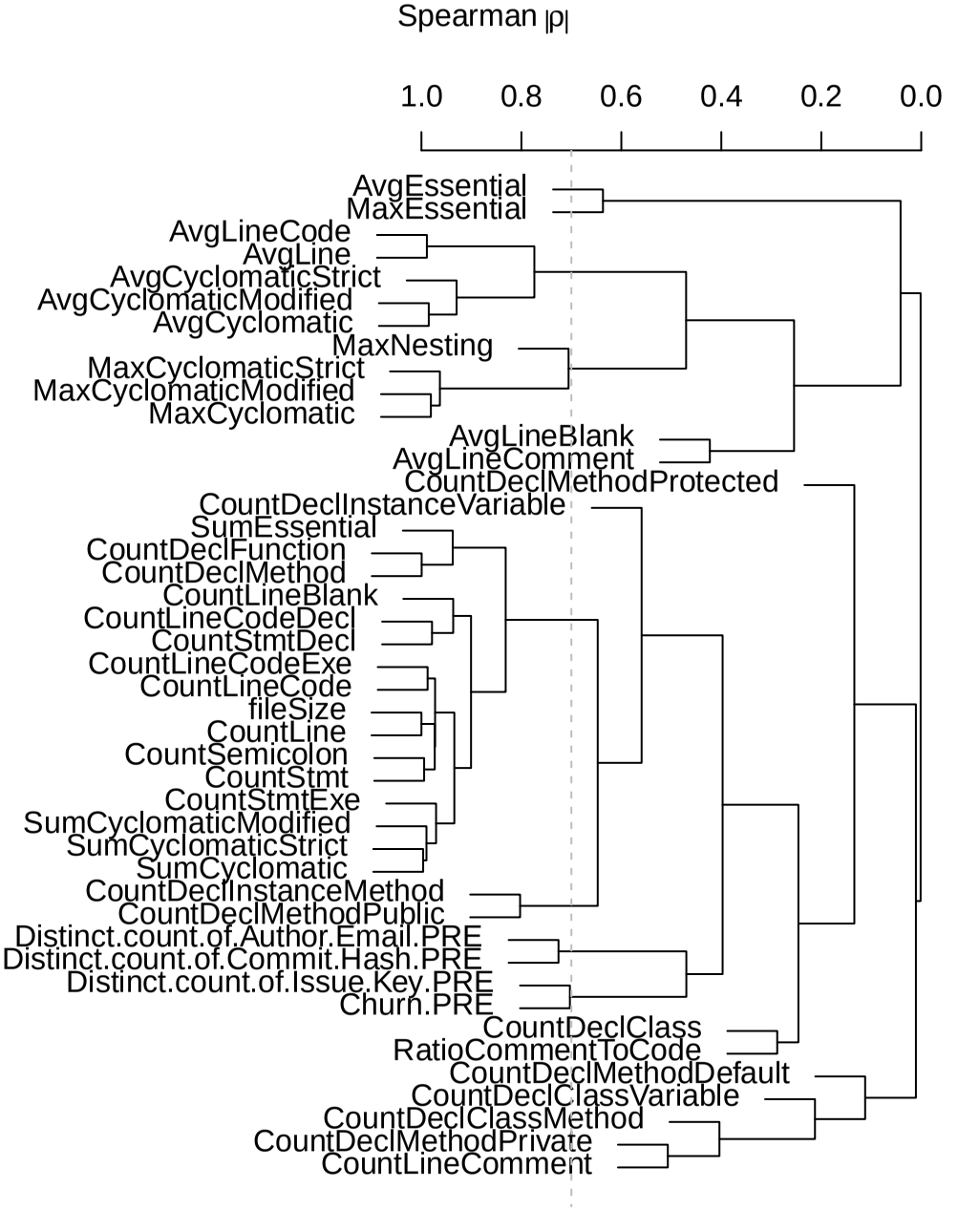

Proportion <= Distinct.count.of.Issue.Key.POST
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Distinct.count.of.Issue.Key.POST

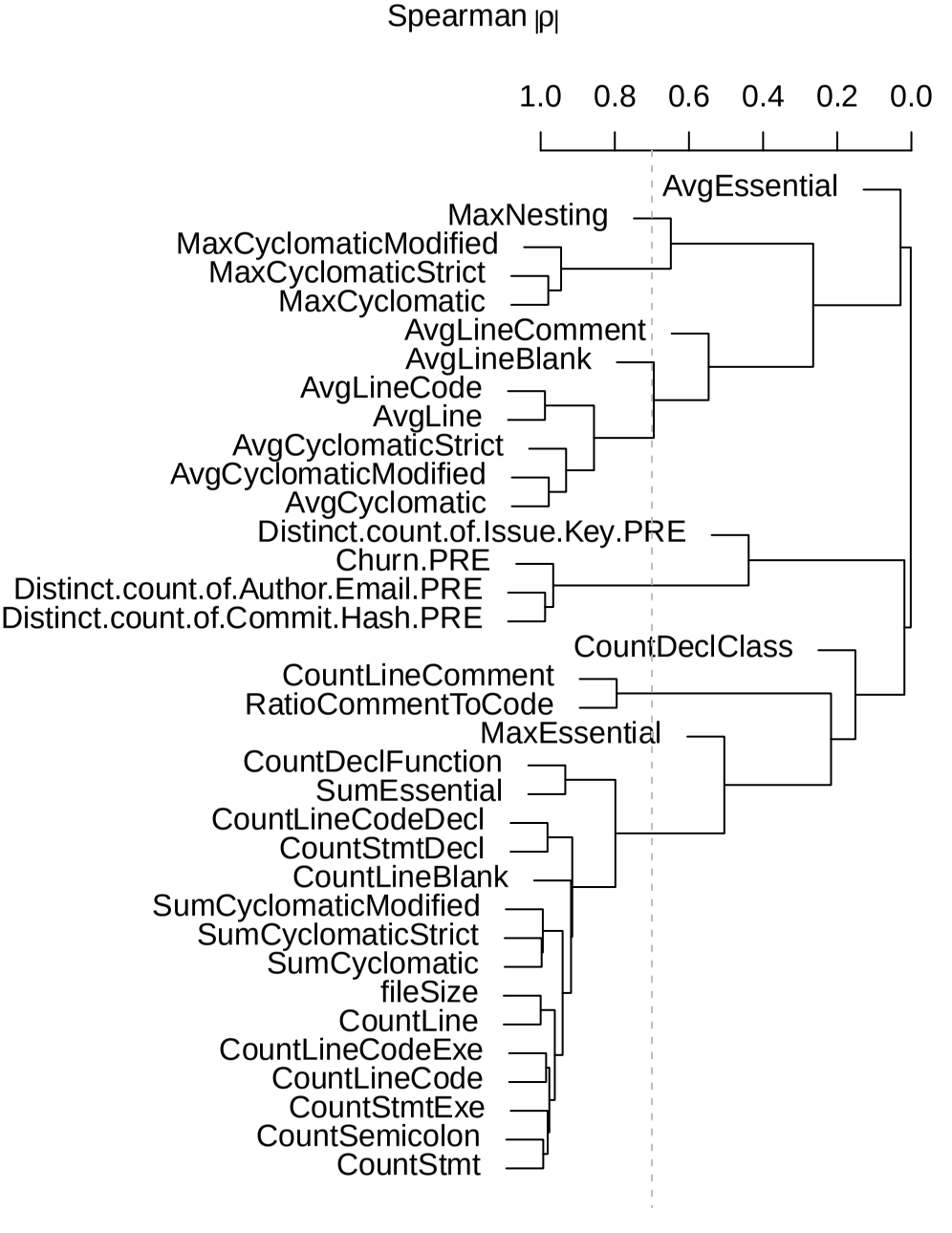

cdf
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.0
0.5
1.0
1.5
2.0
xs

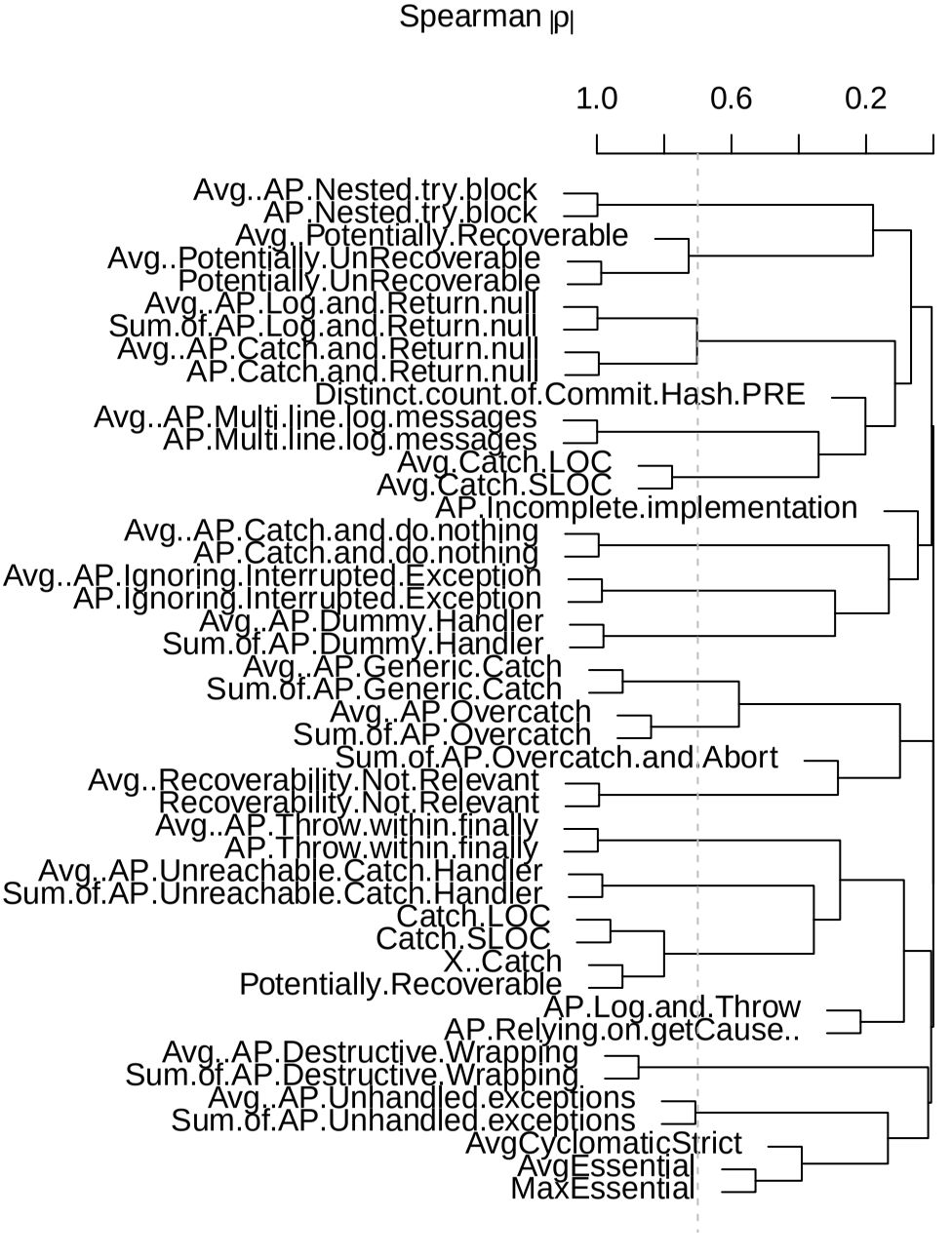

cdf
xs
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0.0
0.2
0.4
0.6
0.8
1.0

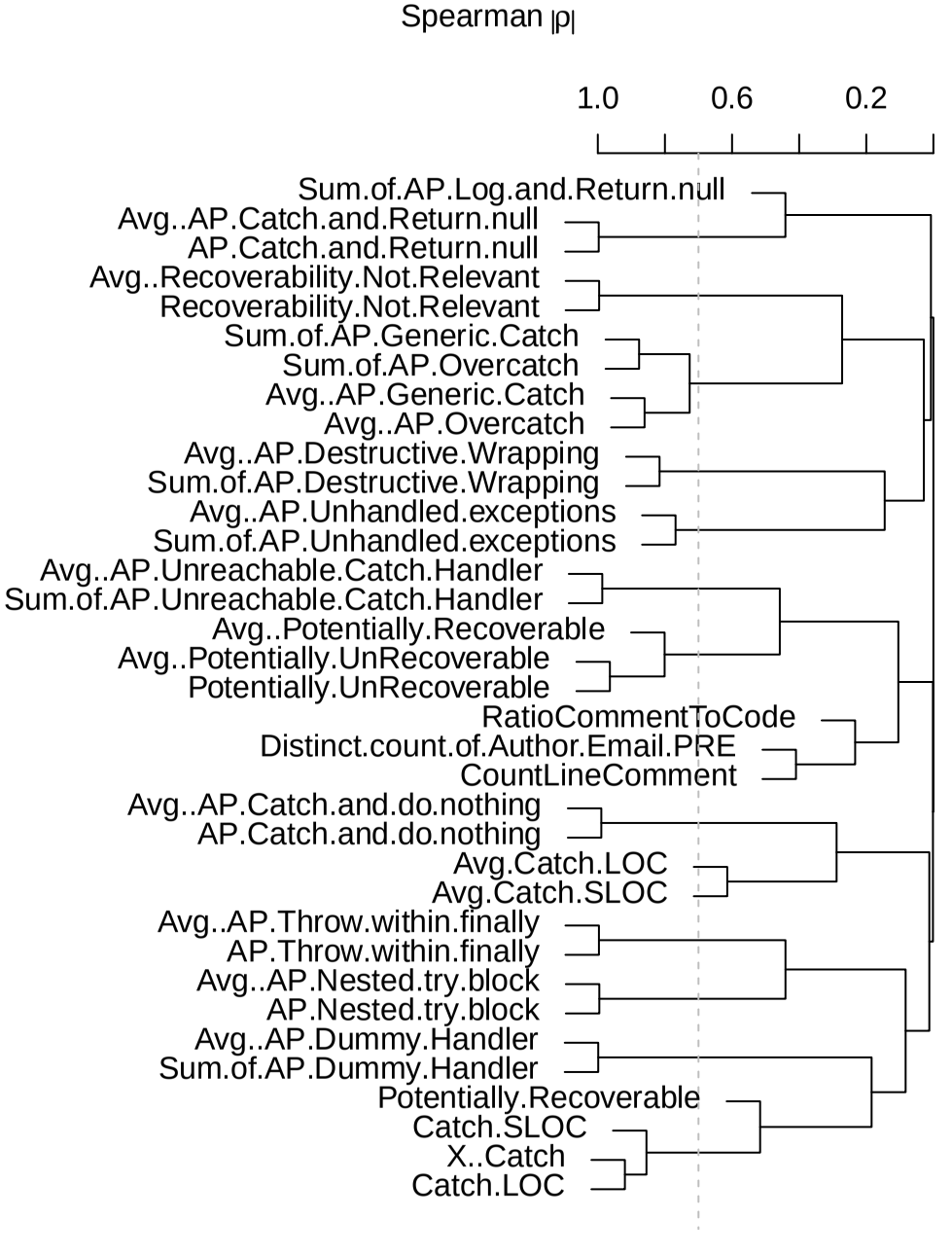

cdf
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.0
0.2
0.4
0.6
0.8
1.0
xs

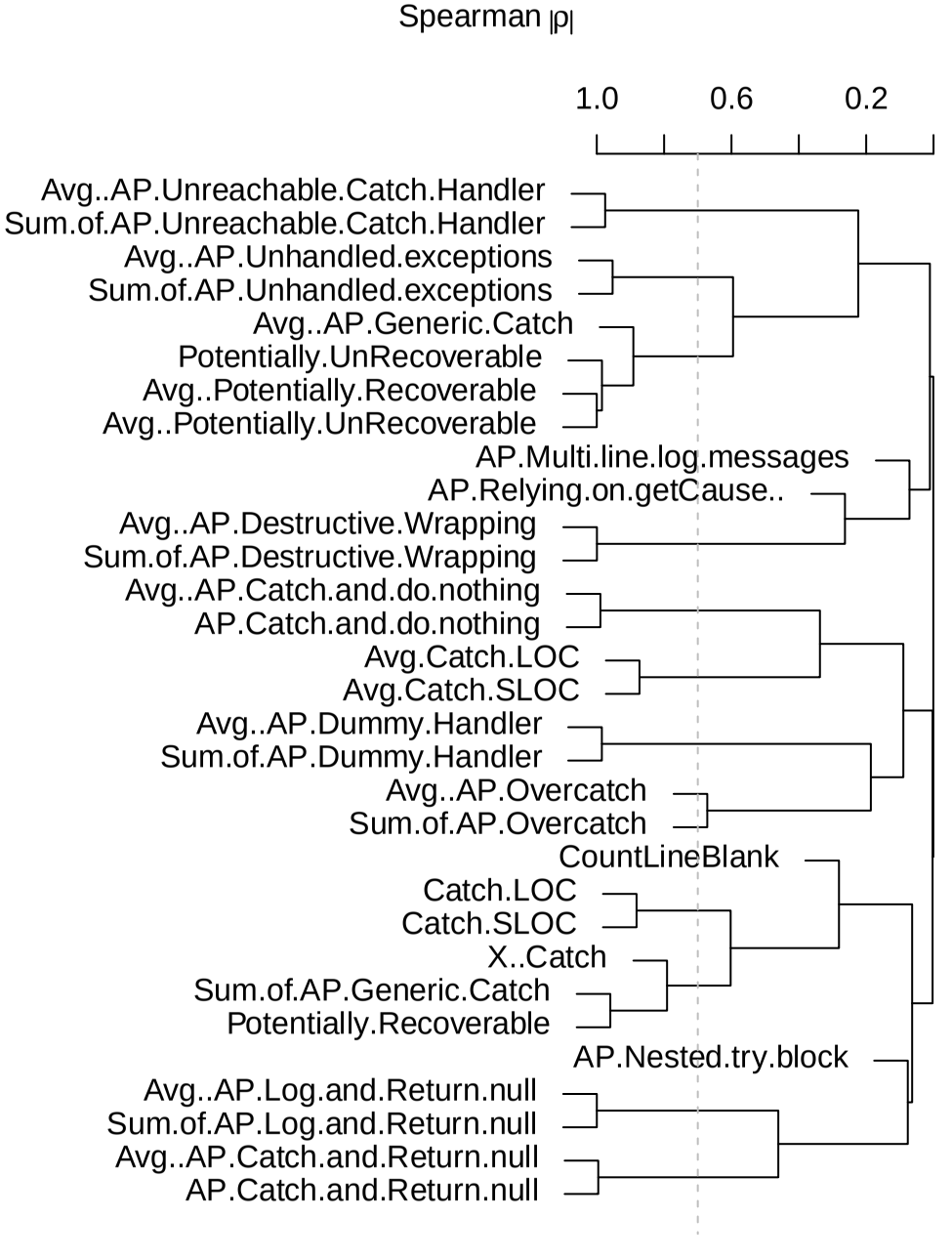

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
AvgEssential
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
AvgLineBlank
AvgLineComment
RatioCommentToCode
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
CountDeclClassMethod
CountDeclClassVariable
CountDeclMethodDefault
MaxEssential
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
CountDeclMethodProtected
CountDeclMethodPublic
CountLineComment
CountDeclMethodPrivate
CountDeclClass
CountDeclInstanceVariable
CountLineCodeDecl
CountStmtDecl
SumCyclomaticModified
SumCyclomaticStrict
SumCyclomatic
fileSize
CountLine
CountStmtExe
CountLineCodeExe
CountLineCode
CountSemicolon
CountStmt
CountDeclInstanceMethod
CountLineBlank
SumEssential
CountDeclFunction
CountDeclMethod

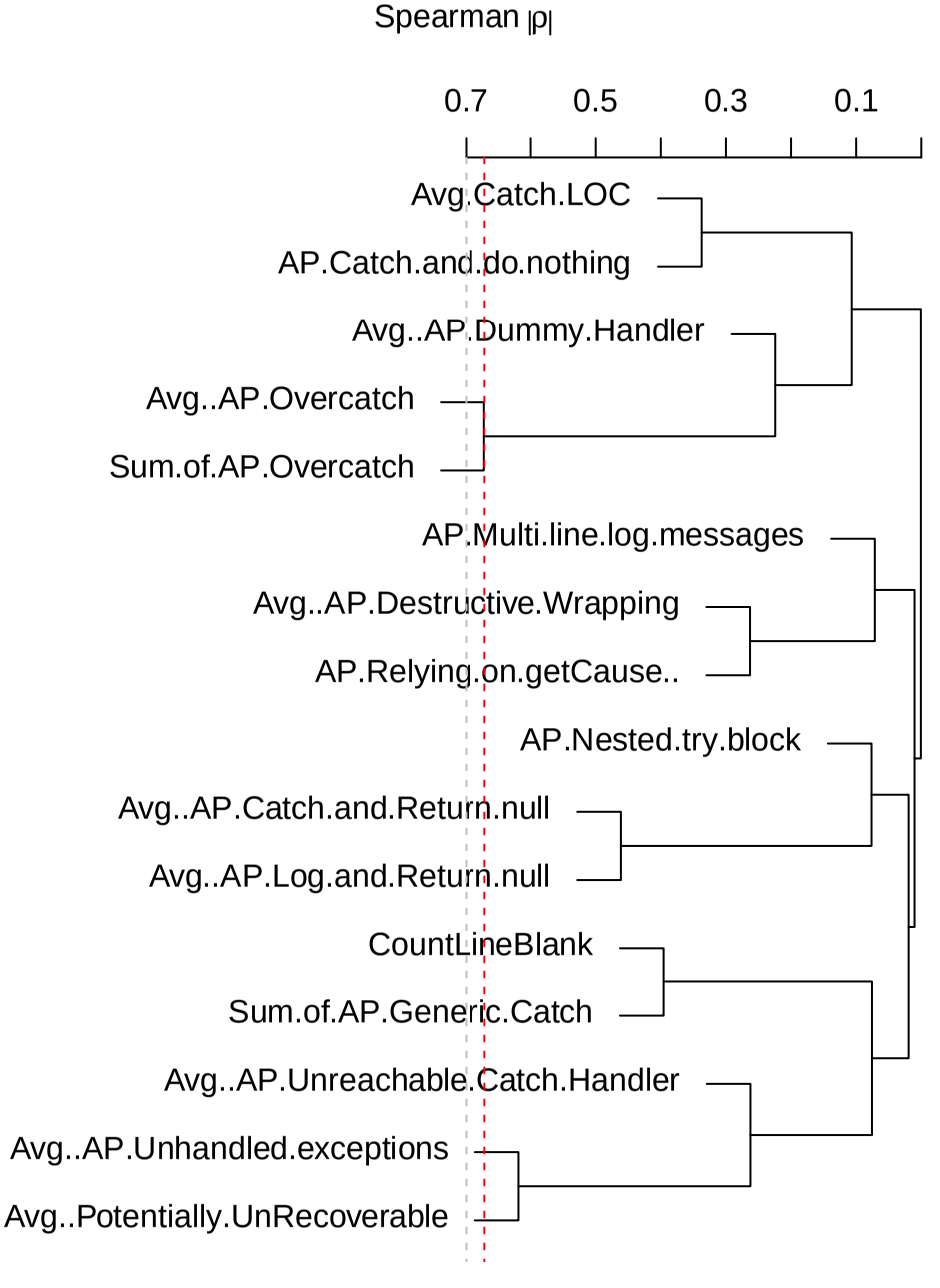

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
AvgEssential
MaxEssential
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
MaxNesting
MaxCyclomaticStrict
MaxCyclomaticModified
MaxCyclomatic
AvgLineBlank
AvgLineComment
CountDeclMethodProtected
CountDeclInstanceVariable
SumEssential
CountDeclFunction
CountDeclMethod
CountLineBlank
CountLineCodeDecl
CountStmtDecl
CountLineCodeExe
CountLineCode
fileSize
CountLine
CountSemicolon
CountStmt
CountStmtExe
SumCyclomaticModified
SumCyclomaticStrict
SumCyclomatic
CountDeclInstanceMethod
CountDeclMethodPublic
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
Distinct.count.of.Issue.Key.PRE
Churn.PRE
CountDeclClass
RatioCommentToCode
CountDeclMethodDefault
CountDeclClassVariable
CountDeclClassMethod
CountDeclMethodPrivate
CountLineComment

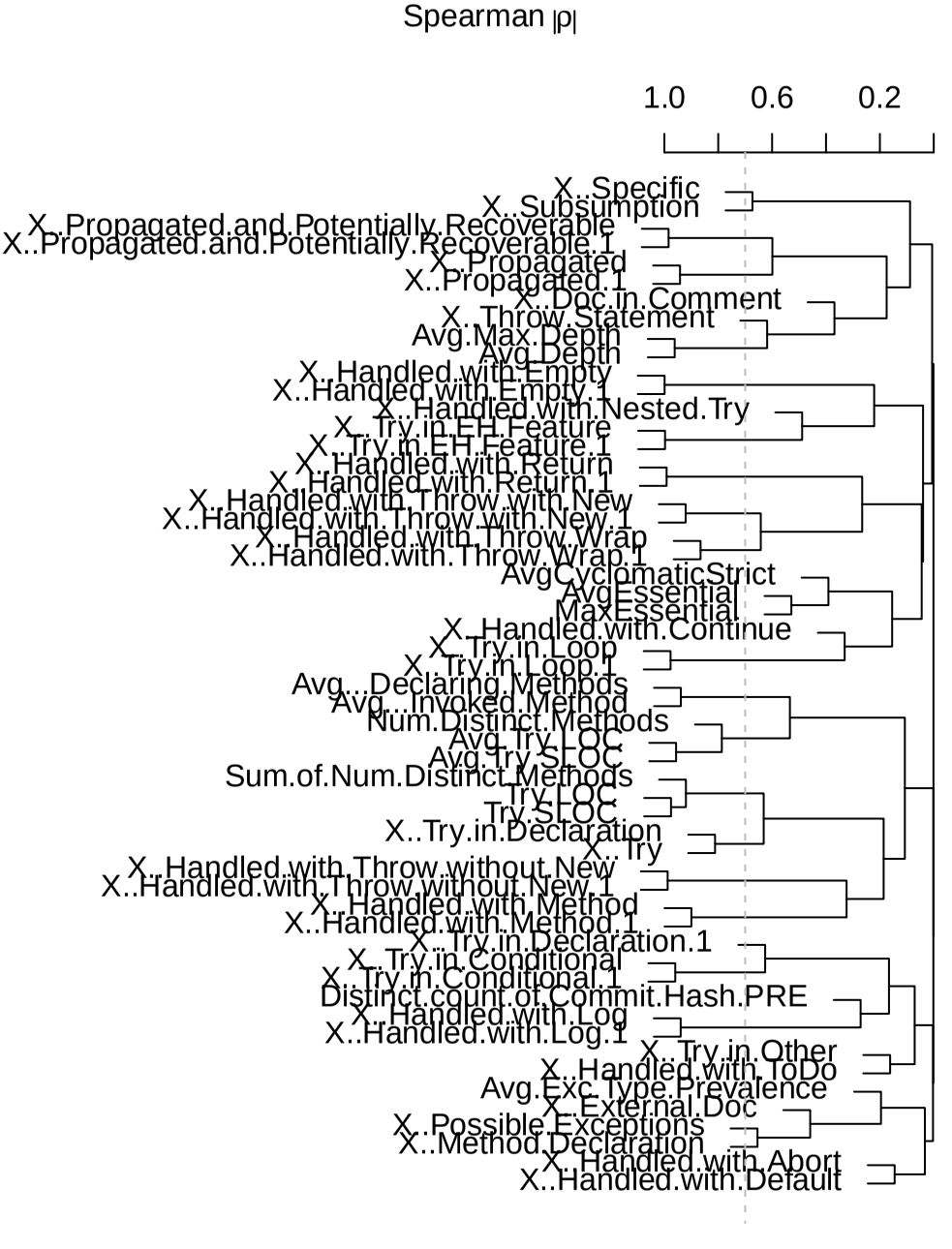

Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
AvgEssential
MaxNesting
MaxCyclomaticModified
MaxCyclomaticStrict
MaxCyclomatic
AvgLineComment
AvgLineBlank
AvgLineCode
AvgLine
AvgCyclomaticStrict
AvgCyclomaticModified
AvgCyclomatic
Distinct.count.of.Issue.Key.PRE
Churn.PRE
Distinct.count.of.Author.Email.PRE
Distinct.count.of.Commit.Hash.PRE
CountDeclClass
CountLineComment
RatioCommentToCode
MaxEssential
CountDeclFunction
SumEssential
CountLineCodeDecl
CountStmtDecl
CountLineBlank
SumCyclomaticModified
SumCyclomaticStrict
SumCyclomatic
fileSize
CountLine
CountLineCodeExe
CountLineCode
CountStmtExe
CountSemicolon
CountStmt

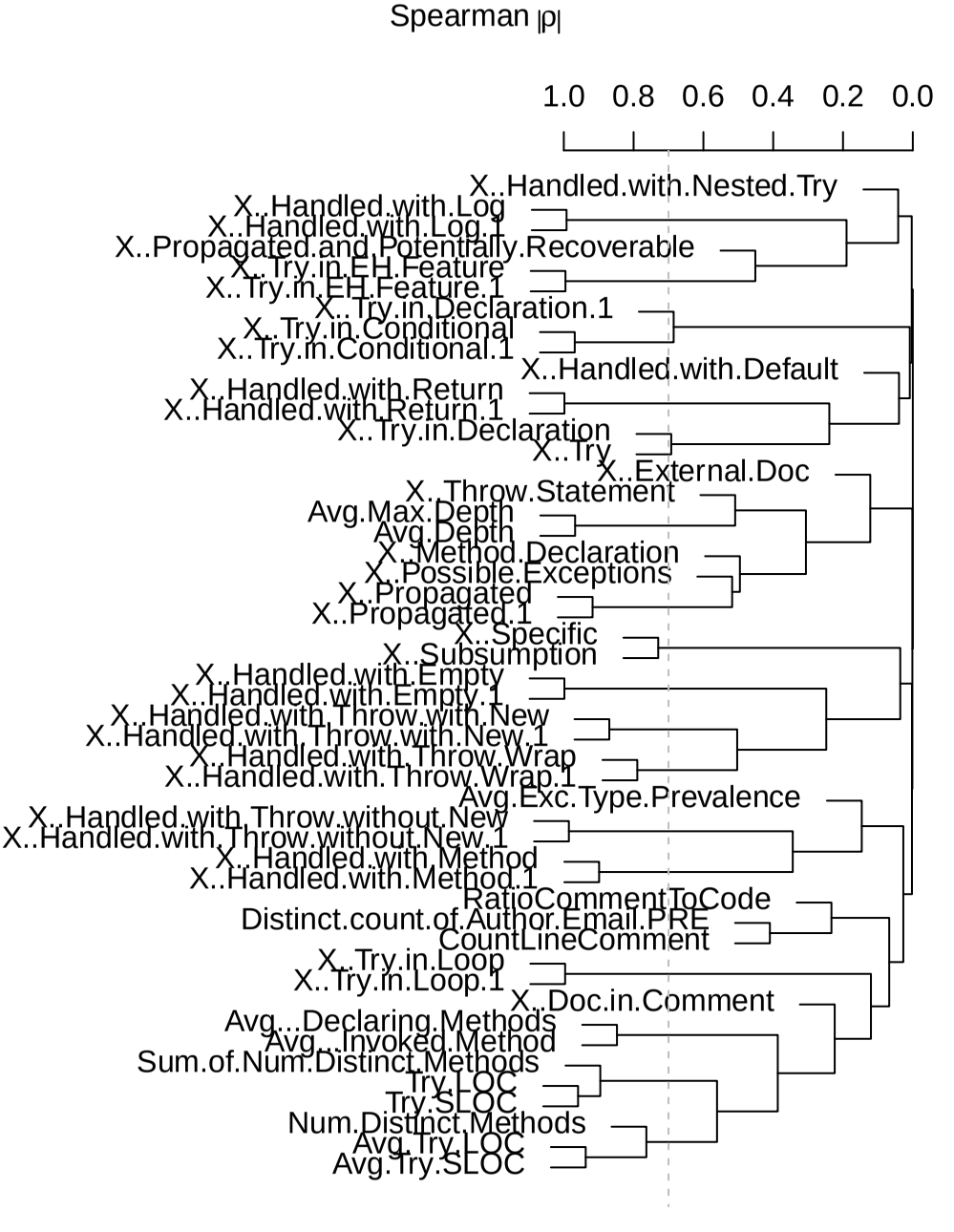

Spearman |ρ|
1.0
0.6
0.2
Avg..AP.Nested.try.block
AP.Nested.try.block
Avg..Potentially.Recoverable
Avg..Potentially.UnRecoverable
Potentially.UnRecoverable
Avg..AP.Log.and.Return.null
Sum.of.AP.Log.and.Return.null
Avg..AP.Catch.and.Return.null
AP.Catch.and.Return.null
Distinct.count.of.Commit.Hash.PRE
Avg..AP.Multi.line.log.messages
AP.Multi.line.log.messages
Avg.Catch.LOC
Avg.Catch.SLOC
AP.Incomplete.implementation
Avg..AP.Catch.and.do.nothing
AP.Catch.and.do.nothing
Avg..AP.Ignoring.Interrupted.Exception
AP.Ignoring.Interrupted.Exception
Avg..AP.Dummy.Handler
Sum.of.AP.Dummy.Handler
Avg..AP.Generic.Catch
Sum.of.AP.Generic.Catch
Avg..AP.Overcatch
Sum.of.AP.Overcatch
Sum.of.AP.Overcatch.and.Abort
Avg..Recoverability.Not.Relevant
Recoverability.Not.Relevant
Avg..AP.Throw.within.finally
AP.Throw.within.finally
Avg..AP.Unreachable.Catch.Handler
Sum.of.AP.Unreachable.Catch.Handler
Catch.LOC
Catch.SLOC
X..Catch
Potentially.Recoverable
AP.Log.and.Throw
AP.Relying.on.getCause..
Avg..AP.Destructive.Wrapping
Sum.of.AP.Destructive.Wrapping
Avg..AP.Unhandled.exceptions
Sum.of.AP.Unhandled.exceptions
AvgCyclomaticStrict
AvgEssential
MaxEssential

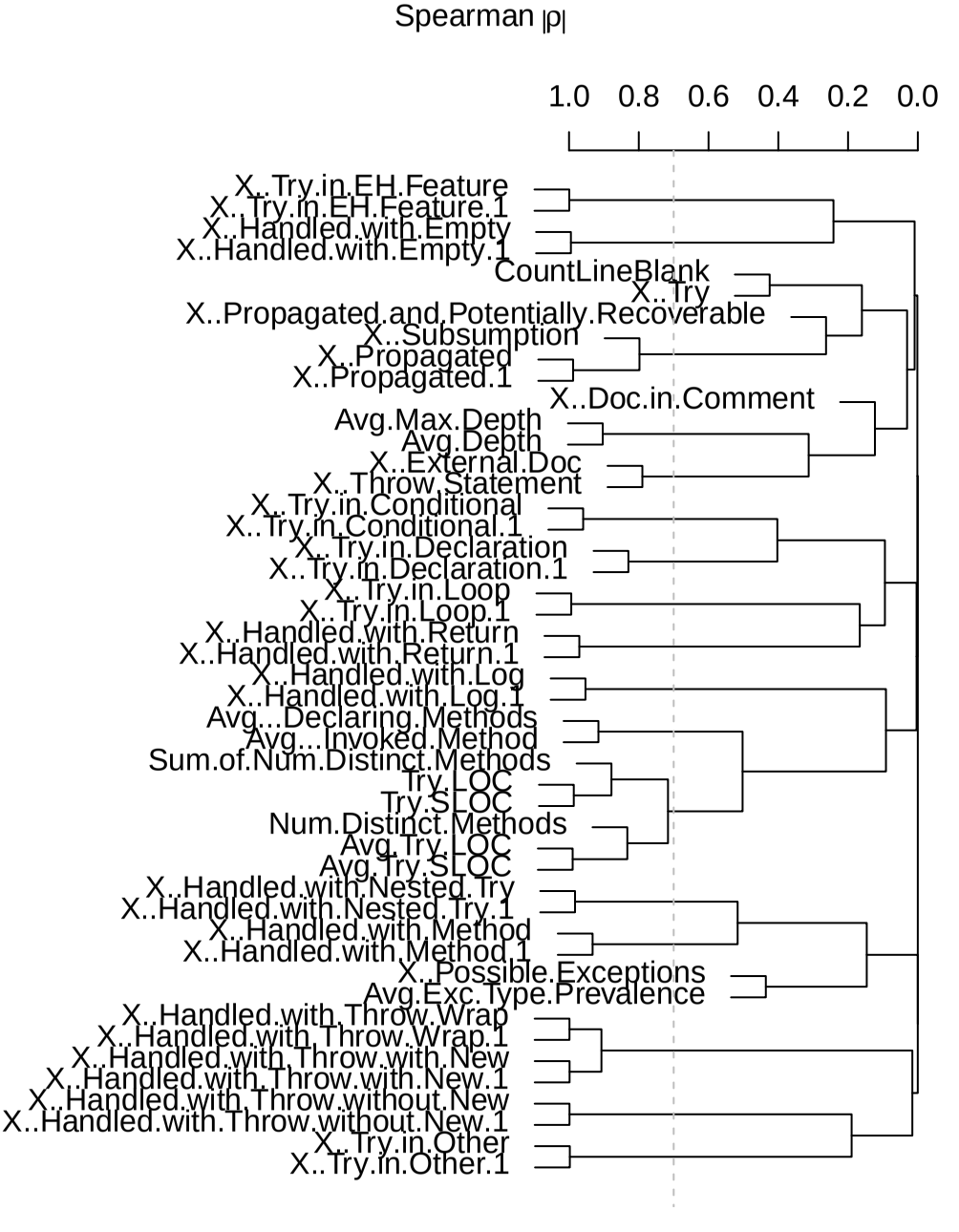

Spearman |ρ|
1.0
0.6
0.2
Sum.of.AP.Log.and.Return.null
Avg..AP.Catch.and.Return.null
AP.Catch.and.Return.null
Avg..Recoverability.Not.Relevant
Recoverability.Not.Relevant
Sum.of.AP.Generic.Catch
Sum.of.AP.Overcatch
Avg..AP.Generic.Catch
Avg..AP.Overcatch
Avg..AP.Destructive.Wrapping
Sum.of.AP.Destructive.Wrapping
Avg..AP.Unhandled.exceptions
Sum.of.AP.Unhandled.exceptions
Avg..AP.Unreachable.Catch.Handler
Sum.of.AP.Unreachable.Catch.Handler
Avg..Potentially.Recoverable
Avg..Potentially.UnRecoverable
Potentially.UnRecoverable
RatioCommentToCode
Distinct.count.of.Author.Email.PRE
CountLineComment
Avg..AP.Catch.and.do.nothing
AP.Catch.and.do.nothing
Avg.Catch.LOC
Avg.Catch.SLOC
Avg..AP.Throw.within.finally
AP.Throw.within.finally
Avg..AP.Nested.try.block
AP.Nested.try.block
Avg..AP.Dummy.Handler
Sum.of.AP.Dummy.Handler
Potentially.Recoverable
Catch.SLOC
X..Catch
Catch.LOC

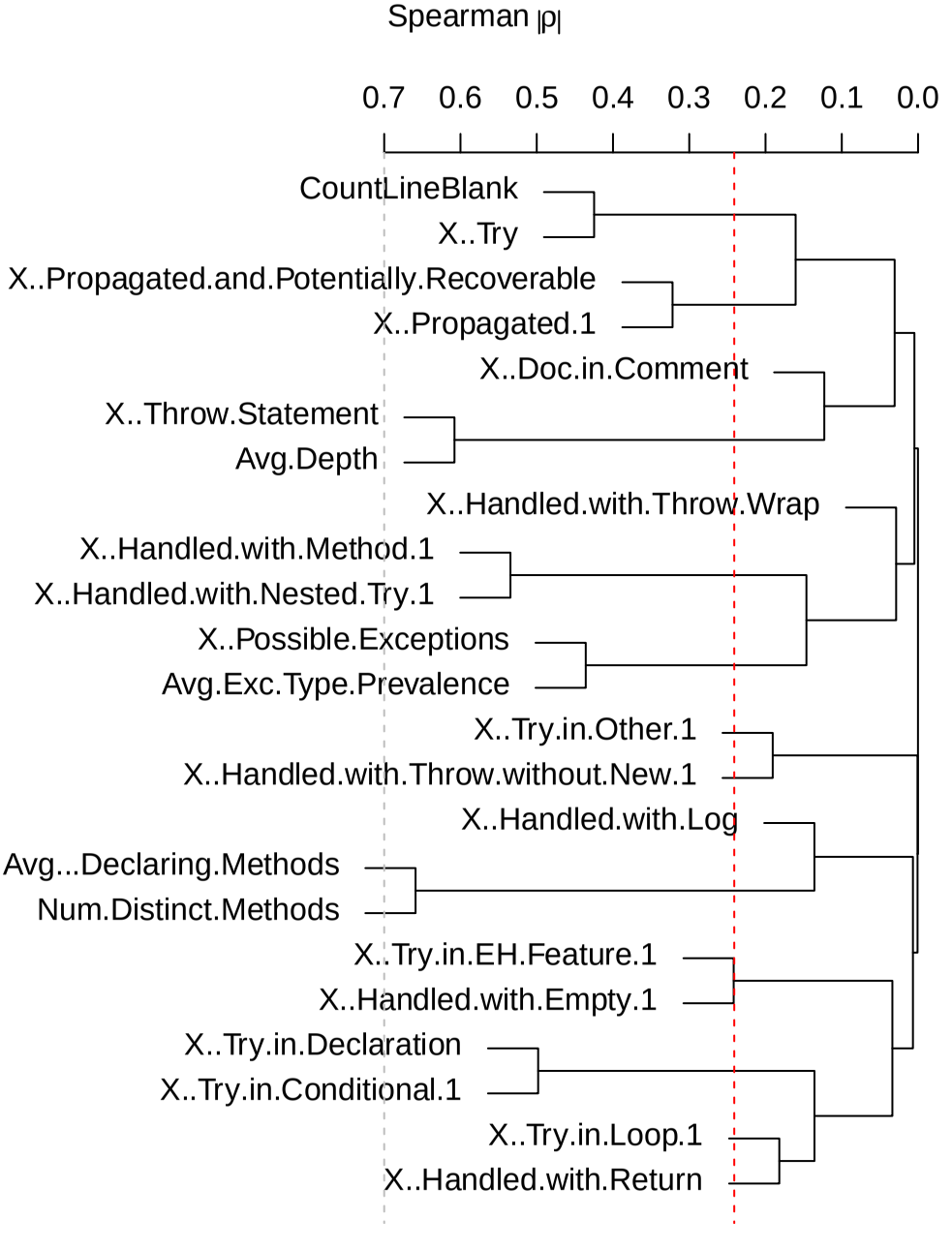

Spearman |ρ|
1.0
0.6
0.2
Avg..AP.Unreachable.Catch.Handler
Sum.of.AP.Unreachable.Catch.Handler
Avg..AP.Unhandled.exceptions
Sum.of.AP.Unhandled.exceptions
Avg..AP.Generic.Catch
Potentially.UnRecoverable
Avg..Potentially.Recoverable
Avg..Potentially.UnRecoverable
AP.Multi.line.log.messages
AP.Relying.on.getCause..
Avg..AP.Destructive.Wrapping
Sum.of.AP.Destructive.Wrapping
Avg..AP.Catch.and.do.nothing
AP.Catch.and.do.nothing
Avg.Catch.LOC
Avg.Catch.SLOC
Avg..AP.Dummy.Handler
Sum.of.AP.Dummy.Handler
Avg..AP.Overcatch
Sum.of.AP.Overcatch
CountLineBlank
Catch.LOC
Catch.SLOC
X..Catch
Sum.of.AP.Generic.Catch
Potentially.Recoverable
AP.Nested.try.block
Avg..AP.Log.and.Return.null
Sum.of.AP.Log.and.Return.null
Avg..AP.Catch.and.Return.null
AP.Catch.and.Return.null


Spearman |ρ|
0.7
0.5
0.3
0.1
Avg.Catch.LOC
AP.Catch.and.do.nothing
Avg..AP.Dummy.Handler
Avg..AP.Overcatch
Sum.of.AP.Overcatch
AP.Multi.line.log.messages
Avg..AP.Destructive.Wrapping
AP.Relying.on.getCause..
AP.Nested.try.block
Avg..AP.Catch.and.Return.null
Avg..AP.Log.and.Return.null
CountLineBlank
Sum.of.AP.Generic.Catch
Avg..AP.Unreachable.Catch.Handler
Avg..AP.Unhandled.exceptions
Avg..Potentially.UnRecoverable


Spearman |ρ|
1.0
0.6
0.2
X..Specific
X..Subsumption
X..Propagated.and.Potentially.Recoverable
X..Propagated.and.Potentially.Recoverable.1
X..Propagated
X..Propagated.1
X..Doc.in.Comment
X..Throw.Statement
Avg.Max.Depth
Avg.Depth
X..Handled.with.Empty
X..Handled.with.Empty.1
X..Handled.with.Nested.Try
X..Try.in.EH.Feature
X..Try.in.EH.Feature.1
X..Handled.with.Return
X..Handled.with.Return.1
X..Handled.with.Throw.with.New
X..Handled.with.Throw.with.New.1
X..Handled.with.Throw.Wrap
X..Handled.with.Throw.Wrap.1
AvgCyclomaticStrict
AvgEssential
MaxEssential
X..Handled.with.Continue
X..Try.in.Loop
X..Try.in.Loop.1
Avg...Declaring.Methods
Avg...Invoked.Method
Num.Distinct.Methods
Avg.Try.LOC
Avg.Try.SLOC
Sum.of.Num.Distinct.Methods
Try.LOC
Try.SLOC
X..Try.in.Declaration
X..Try
X..Handled.with.Throw.without.New
X..Handled.with.Throw.without.New.1
X..Handled.with.Method
X..Handled.with.Method.1
X..Try.in.Declaration.1
X..Try.in.Conditional
X..Try.in.Conditional.1
Distinct.count.of.Commit.Hash.PRE
X..Handled.with.Log
X..Handled.with.Log.1
X..Try.in.Other
X..Handled.with.ToDo
Avg.Exc.Type.Prevalence
X..External.Doc
X..Possible.Exceptions
X..Method.Declaration
X..Handled.with.Abort
X..Handled.with.Default


Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
X..Handled.with.Nested.Try
X..Handled.with.Log
X..Handled.with.Log.1
X..Propagated.and.Potentially.Recoverable
X..Try.in.EH.Feature
X..Try.in.EH.Feature.1
X..Try.in.Declaration.1
X..Try.in.Conditional
X..Try.in.Conditional.1
X..Handled.with.Default
X..Handled.with.Return
X..Handled.with.Return.1
X..Try.in.Declaration
X..Try
X..External.Doc
X..Throw.Statement
Avg.Max.Depth
Avg.Depth
X..Method.Declaration
X..Possible.Exceptions
X..Propagated
X..Propagated.1
X..Specific
X..Subsumption
X..Handled.with.Empty
X..Handled.with.Empty.1
X..Handled.with.Throw.with.New
X..Handled.with.Throw.with.New.1
X..Handled.with.Throw.Wrap
X..Handled.with.Throw.Wrap.1
Avg.Exc.Type.Prevalence
X..Handled.with.Throw.without.New
X..Handled.with.Throw.without.New.1
X..Handled.with.Method
X..Handled.with.Method.1
RatioCommentToCode
Distinct.count.of.Author.Email.PRE
CountLineComment
X..Try.in.Loop
X..Try.in.Loop.1
X..Doc.in.Comment
Avg...Declaring.Methods
Avg...Invoked.Method
Sum.of.Num.Distinct.Methods
Try.LOC
Try.SLOC
Num.Distinct.Methods
Avg.Try.LOC
Avg.Try.SLOC


Spearman |ρ|
1.0
0.8
0.6
0.4
0.2
0.0
X..Try.in.EH.Feature
X..Try.in.EH.Feature.1
X..Handled.with.Empty
X..Handled.with.Empty.1
CountLineBlank
X..Try
X..Propagated.and.Potentially.Recoverable
X..Subsumption
X..Propagated
X..Propagated.1
X..Doc.in.Comment
Avg.Max.Depth
Avg.Depth
X..External.Doc
X..Throw.Statement
X..Try.in.Conditional
X..Try.in.Conditional.1
X..Try.in.Declaration
X..Try.in.Declaration.1
X..Try.in.Loop
X..Try.in.Loop.1
X..Handled.with.Return
X..Handled.with.Return.1
X..Handled.with.Log
X..Handled.with.Log.1
Avg...Declaring.Methods
Avg...Invoked.Method
Sum.of.Num.Distinct.Methods
Try.LOC
Try.SLOC
Num.Distinct.Methods
Avg.Try.LOC
Avg.Try.SLOC
X..Handled.with.Nested.Try
X..Handled.with.Nested.Try.1
X..Handled.with.Method
X..Handled.with.Method.1
X..Possible.Exceptions
Avg.Exc.Type.Prevalence
X..Handled.with.Throw.Wrap
X..Handled.with.Throw.Wrap.1
X..Handled.with.Throw.with.New
X..Handled.with.Throw.with.New.1
X..Handled.with.Throw.without.New
X..Handled.with.Throw.without.New.1
X..Try.in.Other
X..Try.in.Other.1


Spearman |ρ|
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
CountLineBlank
X..Try
X..Propagated.and.Potentially.Recoverable
X..Propagated.1
X..Doc.in.Comment
X..Throw.Statement
Avg.Depth
X..Handled.with.Throw.Wrap
X..Handled.with.Method.1
X..Handled.with.Nested.Try.1
X..Possible.Exceptions
Avg.Exc.Type.Prevalence
X..Try.in.Other.1
X..Handled.with.Throw.without.New.1
X..Handled.with.Log
Avg...Declaring.Methods
Num.Distinct.Methods
X..Try.in.EH.Feature.1
X..Handled.with.Empty.1
X..Try.in.Declaration
X..Try.in.Conditional.1
X..Try.in.Loop.1
X..Handled.with.Return